

ICPC TRD

Main (1), Graphs (1), Trees (9), Range Queries (15), Strings (21), Dynamic Programming (25), Mathematics (27), Geometry (31), Bit Operations (35), Extra (36),

Main

```
// #pragma GCC optimize("O3,unroll-loops")
// #pragma GCC target("avx2,bmi,bmi2,lzcnt,popcnt")

#include <bits/stdc++.h>
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>
using namespace std;
using namespace __gnu_pbds;

#define fastio ios::sync_with_stdio(false); cin.tie(0); cout.tie(0)
#define all(v) v.begin(), v.end()
#define rall(v) v.rbegin(), v.rend()
#define pb push_back
#define fi first
#define se second

typedef long long ll;
typedef long double ld;
typedef pair<int,int> pii;
typedef vector<int> vi;
typedef vector<ll> vll;

const int INF = 1e9;
const ll LINF = 1e18;
const int MOD = 1e9+7;
const double EPS = 1e-9;

void solve(){}

int main() {
    #ifdef ICESI
        freopen("input.txt", "r", stdin);
    #endif
    fastio;
    int t = 1;
    // cin >> t;
    while (t--) {
        solve();
    }
    return 0;
}
```

Graphs

```
/**
 * Uso[1]: dx4, dy4, dx8, dy8, isValid(x, y, R, C);
 * Direcciones de movimiento en grillas 4/8-conexas
 * y validacion de coordenadas dentro de la
 * grilla.
 * Complejidad: O(1).
 */
const int dx4[4] = {1, -1, 0, 0};
const int dy4[4] = {0, 0, 1, -1};
const int dx8[8] = {1, 1, 0, -1, -1, -1, 0, 1};
const int dy8[8] = {0, 1, 1, 1, 0, -1, -1, -1};
#define isValid(x, y, R, C) ((x) >= 0 && (x) < (R)
    && (y) >= 0 && (y) < (C))

/**
 * Uso: Edge<T> e = {from, to, id, weight};
 * Arista con extremos, identificador opcional y
 * peso generico; usada internamente por Graph.
```

```
* Complejidad: O(1) por arista.
*/
template <typename T = ll> struct Edge { int from,
    to, id; T weight; };

/**
 * Uso[0]: DSU dsu(n); dsu.find(x); dsu.unite(a, b);
    dsu.component_size(x); dsu.comps; dsu.
    get_components();
 * Union-find con union por tamaño y acceso a los
    miembros de cada componente.
 * Complejidad: find y unite casi O(α(N)) amortizado
 */
struct DSU {
    vector<int> p, sz;
    vector<vector<int>> members;
    int comps;
    DSU(int n) : p(n), sz(n, 1), members(n), comps(n) {
        iota(all(p), 0);
        for (int i = 0; i < n; i++) members[i].pb(i);
    }
    int find(int x) { return p[x] == x ? x : p[x] =
        find(p[x]); }
    bool unite(int a, int b) {
        a = find(a); b = find(b);
        if (a == b) return false;
        if (sz[a] < sz[b]) swap(a, b);
        p[b] = a; sz[a] += sz[b];
        members[a].insert(members[a].end(), all(members[
            b]));
        vector<int>().swap(members[b]);
        comps--;
        return true;
    }
    const vector<int>& component_nodes(int x) {
        return members[find(x)]; }
    int component_size(int x) { return sz[find(x)]; }
    vector<vector<int>> get_components(int first = 0)
        const {
            vector<vector<int>> res;
            for (int i = first; i < p.size(); i++) if (p[i]
                == i && !members[i].empty()) res.pb(members[i
                ]);
            return res;
        }
};

/**
 * Uso[1]: Graph<ll> g(n); g.add_undirected_edge(u,
    v, w); g.add_directed_edge(u, v, w, id);
 * Grafo con lista de adyacencia de índices a
    aristas y pesos genericos.
 * Complejidad: agregar arista O(1).
 */
template <typename T = ll> struct Graph {
    int n;
    vector<vector<int>> adj; // indices a edges
    vector<Edge<T>> edges;
    Graph(int n) : n(n) { adj.resize(n + 1); }
    void add_directed_edge(int u, int v, T w = 1, int
```

```

        id = -1) {
        adj[u].pb(edges.size());
        edges.pb({u, v, id, w});
    }
    void add_undirected_edge(int u, int v, T w = 1,
        int id = -1) {
        add_directed_edge(u, v, w, id);
        add_directed_edge(v, u, w, id);
    }
};

/**
 * Uso[1]: auto dist = Dijkstra<ll>().run(g, src);
 * auto [dist, parent] = run_with_parents(g, src);
 * Caminos minimos desde un origen con pesos no
 * negativos; opcionalmente reconstruye el camino.
 * Complejidad:  $O(E \log V)$ .
 */
template <typename T = ll> struct Dijkstra {
    //  $O(E \log V)$ 
    vector<T> run(const Graph<T>& g, int src) {
        vector<T> dist(g.n + 1, LINF);
        priority_queue<pair<T, int>, vector<pair<T, int>,
            >>, greater<>> pq;
        dist[src] = 0; pq.push({0, src});
        while (!pq.empty()) {
            auto [d, u] = pq.top(); pq.pop();
            if (d > dist[u]) continue;
            for (int id : g.adj[u]) {
                auto& e = g.edges[id];
                if (dist[u] + e.weight < dist[e.to]) {
                    dist[e.to] = dist[u] + e.weight;
                    pq.push({dist[e.to], e.to});
                }
            }
        }
        return dist;
    }
    // Retorna {distancias, padres} para reconstruir
    // el camino
    pair<vector<T>, vi> run_with_parents(const Graph<
        T>& g, int src) {
        vector<T> dist(g.n + 1, LINF);
        vi p(g.n + 1, -1);
        priority_queue<pair<T, int>, vector<pair<T, int>,
            >>, greater<>> pq;
        dist[src] = 0; pq.push({0, src});
        while (!pq.empty()) {
            auto [d, u] = pq.top(); pq.pop();
            if (d > dist[u]) continue;
            for (int id : g.adj[u]) {
                auto& e = g.edges[id];
                if (dist[u] + e.weight < dist[e.to]) {
                    dist[e.to] = dist[u] + e.weight;
                    p[e.to] = u;
                    pq.push({dist[e.to], e.to});
                }
            }
        }
        return {dist, p};
    }
};

static vi restore_path(int target, const vi& p) {
    vi path;
    for (int v = target; v != -1; v = p[v]) path.pb
        (v);
    reverse(all(path));
}

    return path;
}
};

/**
 * Uso[1]: auto dist = ZeroOneBfs<ll>().run(g, src)
 * ;
 * Caminos minimos con pesos restringidos a {0,1}
 * mediante un deque.
 * Complejidad:  $O(V + E)$ .
 */
template <typename T = ll> struct ZeroOneBfs {
    vector<T> run(const Graph<T>& g, int src) {
        vector<T> dist(g.n + 1, LINF);
        deque<int> dq;
        dist[src] = 0; dq.pb(src);
        while (!dq.empty()) {
            int u = dq.front(); dq.pop_front();
            for (int id : g.adj[u]) {
                auto& e = g.edges[id];
                if (dist[u] + e.weight < dist[e.to]) {
                    dist[e.to] = dist[u] + e.weight;
                    e.weight ? dq.pb(e.to) : dq.push_front(e
                        to);
                }
            }
        }
        return dist;
    }
};

/**
 * Uso[1]: auto [es_bipartito, colores] = Bipartite
 * <ll>().check(g);
 * Verifica si el grafo es bipartito y colorea cada
 * componente con 0/1.
 * Complejidad:  $O(V + E)$ .
 */
template <typename T = ll> struct Bipartite {
    pair<bool, vi> check(const Graph<T>& g) {
        vi color(g.n + 1, -1);
        bool ok = true;
        for (int i = 1; i <= g.n && ok; i++) if (color[
            i] == -1) {
            queue<int> q;
            q.push(i); color[i] = 0;
            while (!q.empty()) {
                int u = q.front(); q.pop();
                for (int id : g.adj[u]) {
                    int v = g.edges[id].to;
                    if (color[v] == -1) color[v] = color[u] ^
                        1, q.push(v);
                    else if (color[v] == color[u]) ok = false
                        ;
                }
            }
        }
        return {ok, color};
    }
};

/**
 * Uso[1]: auto dist = FloydMarshall<ll>().run(g);
 * Caminos minimos entre todos los pares de nodos.
 * Complejidad:  $O(V^3)$  tiempo,  $O(V^2)$  espacio.
 */

```

```

template <typename T = ll> struct FloydWarshall {
    vector<vector<T>> run(const Graph<T>& g) {
        vector<vector<T>> dist(g.n + 1, vector<T>(g.n + 1, LINF));
        for (int i = 1; i <= g.n; i++) dist[i][i] = 0;
        for (auto& e : g.edges) dist[e.from][e.to] = min(dist[e.from][e.to], e.weight);
        for (int k = 1; k <= g.n; k++)
            for (int i = 1; i <= g.n; i++)
                for (int j = 1; j <= g.n; j++)
                    if (dist[i][k] < LINF && dist[k][j] < LINF) dist[i][j] = min(dist[i][j], dist[i][k] + dist[k][j]);
        return dist;
    }
};

/**
 * Uso[1]: auto order = TopoSort<ll>().run(g);
 * Orden topologico de un grafo dirigido aciclico; devuelve vacio si existe un ciclo.
 * Complejidad:  $O(V + E)$ .
 */
template <typename T = ll> struct TopoSort {
    vi run(const Graph<T>& g) {
        vi in(g.n + 1, 0), order;
        for (auto& e : g.edges) in[e.to]++;
        queue<int> q;
        for (int i = 1; i <= g.n; i++) if (!in[i]) q.push(i);
        while (!q.empty()) {
            int u = q.front(); q.pop();
            order.pb(u);
            for (int id : g.adj[u]) if (--in[g.edges[id].to] == 0) q.push(g.edges[id].to);
        }
        return order.size() == g.n ? order : vi();
    }
};

/**
 * Uso[1]: auto comp = TarjanSCC<ll>().run(g);
 * Componentes fuertemente conexas mediante Tarjan; devuelve el id de componente de cada nodo.
 * Complejidad:  $O(V + E)$ .
 */
template <typename T = ll> struct TarjanSCC {
    vi run(const Graph<T>& g) {
        int n = g.n, timer = 0, ncomps = 0;
        vi val(n + 1, 0), comp(n + 1, -1), stk;
        auto dfs = [&](auto&& self, int u) -> int {
            int low = val[u] = ++timer;
            stk.pb(u);
            for (int id : g.adj[u]) {
                int v = g.edges[id].to;
                if (comp[v] < 0) low = min(low, val[v] ? val[v] : self(self, v));
            }
            if (low == val[u]) {
                int x;
                do { x = stk.back(); stk.pop_back(); comp[x] = ncomps; } while (x != u);
                ncomps++;
            }
            return val[u] = low;
        };
        return val;
    }
};

```

```

for (int i = 1; i <= n; i++) if (!val[i]) dfs(dfs, i);
return comp;
}

/**
 * Uso[1]: auto [sin_ciclo_neg, dist, parent, ciclo] = BellmanFord<ll>().run(g, src);
 * Caminos minimos con pesos negativos; detecta y recupera un ciclo negativo si existe.
 * Complejidad:  $O(V \cdot E)$ .
 */
template <typename T = ll> struct BellmanFord {
    tuple<bool, vector<T>, vi, vi> run(const Graph<T>& g, int src) {
        vector<T> dist(g.n + 1, LINF);
        vi p(g.n + 1, -1);
        dist[src] = 0;
        int x = -1;
        for (int i = 0; i < g.n; i++) {
            x = -1;
            for (auto& e : g.edges) if (dist[e.from] < LINF && dist[e.to] > dist[e.from] + e.weight) {
                dist[e.to] = max(-LINF, dist[e.from] + e.weight);
                p[e.to] = e.from; x = e.to;
            }
        }
        vi cycle;
        bool ok = true;
        if (x != -1) {
            ok = false;
            int y = x;
            for (int i = 0; i < g.n; i++) y = p[y];
            for (int cur = y; cur = p[cur]) { cycle.pb(cur); if (cur == y && cycle.size() > 1) break; }
            reverse(all(cycle));
        }
        return {ok, dist, p, cycle};
    }
};

/**
 * Uso[1]: auto ciclo = DirectedCycle<ll>().find(g);
 * Encuentra un ciclo en un grafo dirigido (vacio si no existe).
 * Complejidad:  $O(V + E)$ .
 */
template <typename T = ll> struct DirectedCycle {
    vi find(const Graph<T>& g) {
        int n = g.n;
        vi state(n + 1, 0), parent(n + 1, -1), cycle;
        auto dfs = [&](auto&& self, int u) -> bool {
            state[u] = 1;
            for (int id : g.adj[u]) {
                int v = g.edges[id].to;
                if (state[v] == 0) {
                    parent[v] = u;
                    if (self(self, v)) return true;
                } else if (state[v] == 1) {
                    cycle.pb(v);
                    for (int cur = u; cur != -1 && cur != v; cur = parent[cur]) cycle.pb(cur);
                    cycle.pb(v);
                }
            }
        };
        return cycle;
    }
};

```

```

        reverse(all(cycle));
        return true;
    }
}
state[u] = 2;
return false;
};
for (int i = 1; i <= n; i++) if (!state[i] &&
dfs(dfs, i)) return cycle;
return cycle;
}
};

/**
 * Uso[1]: auto ciclo = UndirectedCycle<ll>().find(g
 * );
 * Encuentra un ciclo en un grafo no dirigido (
 * vacío si no existe).
 * Complejidad:  $O(V + E)$ .
 */
template <typename T = ll> struct UndirectedCycle {
    vi find(const Graph<T>& g) {
        int n = g.n;
        vi state(n + 1, 0), parent(n + 1, -1), cycle;
        auto dfs = [&](auto&& self, int u, int pe) ->
        bool {
            state[u] = 1;
            for (int id : g.adj[u]) {
                if ((id ^ 1) == pe) continue;
                int v = g.edges[id].to;
                if (state[v] == 0) {
                    parent[v] = u;
                    if (self(self, v, id)) return true;
                } else if (state[v] == 1) {
                    cycle.pb(v);
                    for (int cur = u; cur != -1 && cur != v;
                    cur = parent[cur]) cycle.pb(cur);
                    cycle.pb(v);
                    reverse(all(cycle));
                    return true;
                }
            }
            state[u] = 2;
            return false;
        };
        for (int i = 1; i <= n; i++) if (!state[i] &&
        dfs(dfs, i, -1)) return cycle;
        return cycle;
    }
};

/**
 * Uso[1]: auto [peso, aristas_mst] = Kruskal<ll>()
 * .run(g);
 * Arbol de expansion minima (MST) mediante Kruskal
 * ; devuelve el peso total y los indices de las
 * aristas.
 * Complejidad:  $O(E \log E)$ .
 */
template <typename T = ll> struct Kruskal {
    pair<T, vi> run(const Graph<T>& g) {
        vi idx(g.edges.size());
        iota(all(idx), 0);
        sort(all(idx), [&](int a, int b) { return g.
        edges[a].weight < g.edges[b].weight; });
        DSU dsu(g.n + 1);

```

```

        T total = 0;
        vi mst;
        for (int i : idx) {
            auto& e = g.edges[i];
            if (dsu.unite(e.from, e.to)) {
                total += e.weight;
                mst.pb(i);
                if (mst.size() == g.n - 1) break;
            }
        }
        return {total, mst};
    }
};

/**
 * Uso[1]: auto circuito = EulerianCircuit<ll>().
 * find(g, start, undirected);
 * Circuito euleriano dirigido o no dirigido (vacío
 * si no existe).
 * Complejidad:  $O(V + E)$ .
 */
template <typename T = ll> struct EulerianCircuit {
    vi find(const Graph<T>& g, int start = 1, bool
    undirected = true) {
        int n = g.n;
        vi in(n + 1, 0), out(n + 1, 0);
        int max_id = -1;
        for (auto& e : g.edges) {
            out[e.from]++; in[e.to]++;
            max_id = max(max_id, e.id);
        }
        for (int i = 1; i <= n; i++) {
            if (undirected ? (out[i] & 1) : (in[i] != out
            [i])) return {};
        }
        vector<bool> used(max_id + 1, false);
        vi circuit, head(n + 1, 0);
        auto dfs = [&](auto&& self, int u) -> void {
            while (head[u] < g.adj[u].size()) {
                auto& e = g.edges[g.adj[u][head[u]++]];
                if (e.id == -1 || !used[e.id]) {
                    if (e.id != -1) used[e.id] = true;
                    self(self, e.to);
                }
            }
            circuit.pb(u);
        };
        dfs(dfs, start);
        reverse(all(circuit));
        int need = undirected ? g.edges.size() / 2 + 1
        : g.edges.size() + 1;
        if (circuit.size() != need && !g.edges.empty())
            return {};
        return circuit;
    }
};

/**
 * Uso[1]: FunctionalGraph fg(n); fg.build_lifting(
 * succ); fg.kth_successor(u, k); fg.decompose(
 * succ);
 * Grafo funcional (cada nodo un unico sucesor): k-
 * esimo sucesor con binary lifting y
 * descomposicion en ciclos.
 * Complejidad: preproceso  $O(N \log K)$ , k-esimo
 * sucesor  $O(\log K)$ .

```

```

*/
struct FunctionalGraph {
    int n, log_k;
    vector<vector<int>> up;
    vi succ, in_cycle, cycle_id, cycle_pos,
        cycle_size, dist_to_cycle;
    vector<vi> cycles;
    FunctionalGraph(int n, int log_k = 60) : n(n),
        log_k(log_k),
        up(n + 1, vector<int>(log_k, 0)), succ(n + 1,
        0), in_cycle(n + 1, 0),
        cycle_id(n + 1, -1), cycle_pos(n + 1, -1),
        cycle_size(n + 1, 0), dist_to_cycle(n + 1, 0)
    {}
    // Descompone el grafo en ciclos y arboles
    // colgantes
    void decompose(const vi& _succ) {
        succ = _succ;
        vi state(n + 1, 0);
        cycles.clear();
        for (int i = 1; i <= n; i++) if (!state[i]) {
            int cur = i;
            vi path;
            while (cur >= 1 && cur <= n && state[cur] ==
            0) {
                state[cur] = 1;
                path.pb(cur);
                cur = succ[cur];
            }
            if (cur >= 1 && cur <= n && state[cur] == 1)
            {
                vi cyc;
                bool start = false;
                for (int u : path) { if (u == cur) start =
                true; if (start) cyc.pb(u); }
                int cid = cycles.size(), csz = cyc.size();
                for (int pos = 0; pos < csz; pos++) {
                    int u = cyc[pos];
                    in_cycle[u] = 1; cycle_id[u] = cid;
                    cycle_pos[u] = pos; cycle_size[u] = csz;
                    dist_to_cycle[u] = 0;
                }
                cycles.pb(cyc);
            }
            for (int u : path) state[u] = 2;
        }
        for (int i = 1; i <= n; i++) if (!in_cycle[i])
        {
            int cur = i;
            vi path;
            while (cur >= 1 && cur <= n && !in_cycle[cur]
            && cycle_id[cur] == -1) path.pb(cur), cur =
            succ[cur];
            int cid = (cur >= 1 && cur <= n) ? cycle_id[
            cur] : -1;
            int csz = (cur >= 1 && cur <= n) ? cycle_size
            [cur] : 0;
            int d = (cur >= 1 && cur <= n) ?
            dist_to_cycle[cur] : 0;
            for (int j = path.size() - 1; j >= 0; j--) {
                int u = path[j];
                cycle_id[u] = cid; cycle_size[u] = csz;
                dist_to_cycle[u] = ++d;
            }
        }
    }
}

```

```

// Binary lifting sobre succ para responder k-
// esimo sucesor
void build_lifting(const vi& _succ) {
    succ = _succ;
    for (int i = 1; i <= n; i++) up[i][0] = succ[i];
    for (int j = 1; j < log_k; j++)
        for (int i = 1; i <= n; i++) {
            int p = up[i][j - 1];
            up[i][j] = (p >= 1 && p <= n) ? up[p][j -
            1] : 0;
        }
    int kth_successor(int u, ll k) {
        for (int j = 0; j < log_k; j++) if (k & (1LL <<
        j)) {
            u = up[u][j];
            if (u < 1 || u > n) return 0;
        }
        return u;
    }
};

/**
 * Uso[1]: LCA lca(n); lca.build(root, adj); lca.
 * lca(u, v); lca.dist(u, v);
 * Ancestro comun mas bajo y distancia en numero de
 * aristas mediante binary lifting.
 * Complejidad: preproceso O(N log N), consulta
 * O(log N).
 */
struct LCA {
    int n, log_n;
    vector<vi> up;
    vi depth;
    LCA(int n, int log_n = 20) : n(n), log_n(log_n),
        up(n + 1, vi(log_n, 0)), depth(n + 1, 0) {}
    void dfs(int u, int p, const vector<vi>& adj) {
        up[u][0] = p;
        for (int j = 1; j < log_n; j++) up[u][j] = up[
        up[u][j - 1]][j - 1];
        for (int v : adj[u]) if (v != p) depth[v] =
        depth[u] + 1, dfs(v, u, adj);
    }
    void build(int root, const vector<vi>& adj) { dfs
    (root, root, adj); }
    int lca(int u, int v) {
        if (depth[u] < depth[v]) swap(u, v);
        int k = depth[u] - depth[v];
        for (int j = 0; j < log_n; j++) if (k & (1 << j
        )) u = up[u][j];
        if (u == v) return u;
        for (int j = log_n - 1; j >= 0; j--) if (up[u][
        j] != up[v][j]) u = up[u][j], v = up[v][j];
        return up[u][0];
    }
    int dist(int u, int v) { return depth[u] + depth[
    v] - 2 * depth[lca(u, v)]; }
};

/**
 * Uso[1]: TwoSat s(n); s.add_clause(u, tu, v, tv);
 * s.force_value(u, t); s.solve();
 * Satisfacibilidad booleana 2-SAT con clausulas (
 * or, implicacion, xor, equivalencia) y la
 * asignacion resultante.
 */

```

```

* Complejidad:  $O(V + E)$ .
*/
struct TwoSat {
    int n;
    Graph<ll> G;
    vector<bool> assignment;
    TwoSat(int n) : n(n), G(2 * n), assignment(n + 1,
        false) {}
    int node(int u, bool t) { return t ? u : u + n; }
    void add_clause(int u, bool tu, int v, bool tv) {
        G.add_directed_edge(node(u, !tu), node(v, tv));
        G.add_directed_edge(node(v, !tv), node(u, tu));
    }
    void force_value(int u, bool t) { add_clause(u, t,
        u, t); }
    void add_implication(int u, bool tu, int v, bool
        tv) { add_clause(u, !tu, v, tv); }
    void add_xor(int u, bool tu, int v, bool tv) {
        add_clause(u, tu, v, tv);
        add_clause(u, !tu, v, !tv);
    }
    void add_equivalence(int u, bool tu, int v, bool
        tv) {
        add_implication(u, tu, v, tv);
        add_implication(v, tv, u, tu);
    }
    bool solve() {
        vi comp = TarjanSCC<ll>().run(G);
        for (int i = 1; i <= n; i++) {
            if (comp[i] == comp[i + n]) return false;
            assignment[i] = comp[i] < comp[i + n];
        }
        return true;
    }
};

/**
* Uso[1]: BipartiteMatcher bm(n_izq, n_der); bm.
    add_edge(u, v); bm.max_matching();
* Emparejamiento maximo bipartito con el algoritmo
  de Kuhn.
* Complejidad:  $O(V \cdot E)$ .
*/
struct BipartiteMatcher {
    int n, m;
    vector<vi> adj;
    vi match;
    BipartiteMatcher(int n, int m) : n(n), m(m), adj(
        n + 1), match(m + 1, -1) {}
    void add_edge(int u, int v) { adj[u].pb(v); }
    bool dfs(int u, vector<bool>& vis) {
        for (int v : adj[u]) if (!vis[v]) {
            vis[v] = true;
            if (match[v] < 0 || dfs(match[v], vis)) {
                match[v] = u; return true; }
        }
        return false;
    }
    int max_matching() {
        int ans = 0;
        for (int i = 1; i <= n; i++) {
            vector<bool> vis(m + 1, false);
            if (dfs(i, vis)) ans++;
        }
        return ans;
    }
};

```

```

};

/**
* Uso[1]: HopcroftKarp hk(n_izq, n_der); hk.
    add_edge(u, v); hk.max_matching();
* Emparejamiento maximo bipartito con el algoritmo
  de Hopcroft-Karp.
* Complejidad:  $O(E\sqrt{V})$ .
*/
struct HopcroftKarp {
    int n, m;
    vector<vi> adj;
    vi pairU, pairV, dist;
    HopcroftKarp(int n, int m) : n(n), m(m), adj(n +
        1), pairU(n + 1, 0), pairV(m + 1, 0), dist(n +
        1) {}
    void add_edge(int u, int v) { adj[u].pb(v); }
    bool bfs() {
        queue<int> q;
        for (int u = 1; u <= n; u++) {
            if (pairU[u]) dist[u] = INF;
            else dist[u] = 0, q.push(u);
        }
        dist[0] = INF;
        while (!q.empty()) {
            int u = q.front(); q.pop();
            if (dist[u] < dist[0])
                for (int v : adj[u]) if (dist[pairV[v]] ==
                    INF) dist[pairV[v]] = dist[u] + 1, q.push(pairV[
                    v]);
        }
        return dist[0] != INF;
    }
    bool dfs(int u) {
        if (!u) return true;
        for (int v : adj[u]) if (dist[pairV[v]] == dist
            [u] + 1 && dfs(pairV[v])) {
            pairV[v] = u; pairU[u] = v; return true;
        }
        dist[u] = INF;
        return false;
    }
    int max_matching() {
        int res = 0;
        while (bfs()) for (int u = 1; u <= n; u++) if
            (!pairU[u] && dfs(u)) res++;
        return res;
    }
};

/**
* Uso[0]: MaxClique mc(n); mc.add_edge(u, v); mc.
    solve();
* Clique maxima de un grafo con hasta 60 nodos
  usando bitsets.
* Complejidad: exponencial en el peor caso,
  practico para  $N \leq 60$ .
*/
struct MaxClique {
    int n, max_sz;
    vector<ll> adj;
    MaxClique(int n) : n(n), max_sz(0), adj(n, 0) {}
    void add_edge(int u, int v) { adj[u] |= (1LL << v
        ); adj[v] |= (1LL << u); }
    void dfs(ll R, ll P, ll X) {
        if (!P && !X) { max_sz = max(max_sz,
            __builtin_popcountll(R)); return; }
    }
};

```

```

    ll iter = P & ~adj[__builtin_ctzll(P | X)];
    while (iter) {
        int v = __builtin_ctzll(iter); ll v_bit = 1LL
        << v;
        dfs(R | v_bit, P & adj[v], X & adj[v]);
        P &= ~v_bit; X |= v_bit; iter &= ~v_bit;
    }
}
int solve() { dfs(0, (1LL << n) - 1, 0); return
max_sz; }
};
/**
 * Uso[1]: BridgeTree bt(n, num_aristas); bt.
 * add_edge(u, v, id); bt.build();
 * Detecta puentes y puntos de articulacion y
 * construye el arbol de componentes 2-arista-
 * conexas.
 * Complejidad:  $O(V + E)$ .
 */
struct BridgeTree {
    int n, timer, num_comps;
    vector<vector<pair<int, int>>> adj;
    vi tin, low, comp_id;
    vector<bool> is_bridge, is_articulation;
    vector<vi> tree_adj;
    BridgeTree(int n, int num_edges) : n(n), adj(n +
    1), tin(n + 1, -1), low(n + 1, -1),
    comp_id(n + 1, 0), is_bridge(num_edges, false),
    is_articulation(n + 1, false), timer(0),
    num_comps(0) {}
    void add_edge(int u, int v, int id) { adj[u].pb({
    v, id}); adj[v].pb({u, id}); }
    void dfs_tarjan(int u, int p = -1) {
        tin[u] = low[u] = ++timer;
        int children = 0;
        for (auto [v, id] : adj[u]) {
            if (v == p) continue;
            if (tin[v] != -1) low[u] = min(low[u], tin[v
            ]);
            else {
                children++;
                dfs_tarjan(v, u);
                low[u] = min(low[u], low[v]);
                if (low[v] > tin[u]) is_bridge[id] = true;
                if (low[v] >= tin[u] && p != -1)
                    is_articulation[u] = true;
            }
        }
        if (p == -1 && children > 1) is_articulation[u]
        = true;
    }
    void dfs_comp(int u, int cur) {
        comp_id[u] = cur;
        for (auto [v, id] : adj[u]) if (!comp_id[v]) {
            if (is_bridge[id]) {
                tree_adj.pb(vi());
                tree_adj[cur].pb(++num_comps);
                tree_adj[num_comps].pb(cur);
                dfs_comp(v, num_comps);
            } else dfs_comp(v, cur);
        }
    }
    void build() {
        for (int i = 1; i <= n; i++) if (tin[i] == -1)
        dfs_tarjan(i);
        tree_adj.pb(vi());

```

```

        for (int i = 1; i <= n; i++) if (!comp_id[i])
        tree_adj.pb(vi()), dfs_comp(i, ++num_comps);
    }
};
/**
 * Uso[0]: Dinic d(n, s, t); d.add_edge(u, v, cap);
 * d.max_flow();
 * Flujo maximo con el algoritmo de Dinic, dirigido
 * o no dirigido.
 * Complejidad:  $O(V^2E)$  en general,  $O(E\sqrt{V})$  en redes
 * unitarias.
 */
struct Dinic {
    struct FlowEdge { int u, v; ll cap, flow = 0; };
    int n, s, t;
    vector<FlowEdge> edges;
    vector<vi> adj;
    vi level, ptr;
    Dinic(int n, int s, int t) : n(n), s(s), t(t),
    adj(n + 1), level(n + 1), ptr(n + 1) {}
    int add_edge(int u, int v, ll cap, bool directed
    = true) {
        adj[u].pb(edges.size()); edges.pb({u, v, cap});
        adj[v].pb(edges.size()); edges.pb({v, u,
        directed ? 0 : cap});
        return (int)edges.size() - 2; // indice de la
        arista forward
    }
    bool bfs() {
        fill(all(level), -1);
        level[s] = 0;
        queue<int> q;
        q.push(s);
        while (!q.empty()) {
            int u = q.front(); q.pop();
            for (int id : adj[u]) {
                auto& e = edges[id];
                if (e.cap - e.flow >= 1 && level[e.v] ==
                -1) level[e.v] = level[u] + 1, q.push(e.v);
            }
        }
        return level[t] != -1;
    }
    ll dfs(int u, ll pushed) {
        if (!pushed) return 0;
        if (u == t) return pushed;
        for (int& cid = ptr[u]; cid < adj[u].size();
        cid++) {
            auto& e = edges[adj[u][cid]];
            ll tr = e.cap - e.flow;
            if (level[u] + 1 != level[e.v] || !tr)
                continue;
            ll push = dfs(e.v, min(pushed, tr));
            if (push) {
                e.flow += push;
                edges[adj[u][cid] ^ 1].flow -= push;
                return push;
            }
        }
        return 0;
    }
    ll max_flow() { return max_flow(s, t); }
    ll max_flow(int src, int snk) {
        s = src; t = snk;
        ll flow = 0;

```

```

    while (bfs()) {
        fill(all(ptr), 0);
        while (ll pushed = dfs(s, LINF)) flow +=
            pushed;
    }
    return flow;
}

/**
 * Uso[1]: DinicLB d(n); int id = d.add_edge(u, v,
 * lo, hi); ll flujo = d.max_flow(s, t); ll f = d.
 * edge_flow(id);
 * Flujo maximo con cotas inferiores (lower bounds)
 * en cada arista; devuelve -1 si no existe flujo
 * factible.
 * Complejidad: O(Dinic) sobre el grafo
 * transformado (dos fases de flujo maximo).
 */
struct DinicLB {
    int n, ss, tt;
    Dinic din;
    vector<ll> bal, lo;
    vector<int> edge_idx;
    bool built = false;
    ll total = 0;
    DinicLB(int n) : n(n), ss(0), tt(n + 1), din(n +
        2, 0, 0), bal(n + 2, 0) {}
    int add_edge(int u, int v, ll lower, ll upper) {
        bal[u] -= lower; bal[v] += lower;
        lo.pb(lower);
        int id = din.add_edge(u, v, upper - lower);
        edge_idx.pb(id);
        return (int)lo.size() - 1;
    }
    void build_super() {
        if (built) return;
        built = true;
        for (int i = 1; i <= n; i++) {
            if (bal[i] > 0) din.add_edge(ss, i, bal[i]),
                total += bal[i];
            else if (bal[i] < 0) din.add_edge(i, tt, -bal
                [i]);
        }
    }
    bool feasible() {
        build_super();
        return din.max_flow(ss, tt) == total;
    }
    ll max_flow(int s, int t) {
        int idx = din.add_edge(t, s, LINF);
        if (!feasible()) return -1;
        ll flow = din.edges[idx].flow; // flujo de la
        // circulacion (valor base s->t)
        // elimina t->s [cap y flow en ambas direcciones
        // , si no queda flujo residual cancelable]
        din.edges[idx].flow = 0; din.edges[idx].cap =
            0;
        din.edges[idx ^ 1].flow = 0; din.edges[idx ^
            1].cap = 0;
        return flow + din.max_flow(s, t);
    }
    ll edge_flow(int id) const { return lo[id] + din.
        edges[edge_idx[id]].flow; }
};

```

```

/**
 * Uso[0]: MCMF m(n); m.add_edge(u, v, cap, costo);
 * auto [flujo, costo] = m.solve(s, t);
 * Flujo maximo de costo minimo (min cost max flow)
 * con potenciales para costos negativos.
 * Complejidad: O(F · E log V) con F el flujo total.
 */
struct MCMF {
    struct Edge { int to, rev; ll cap, flow, cost; };
    int n;
    vector<vector<Edge>> adj;
    vll dist, pot;
    vi p_node, p_edge;
    MCMF(int n) : n(n), adj(n + 1), dist(n + 1), pot(
        n + 1, 0), p_node(n + 1), p_edge(n + 1) {}
    void add_edge(int u, int v, ll cap, ll cost) {
        adj[u].pb({v, (int)adj[v].size(), cap, 0, cost
        });
        adj[v].pb({u, (int)adj[u].size() - 1, 0, 0, -
        cost});
    }
    // Inicializa potenciales si hay costos negativos
    // . Omitir si todos son >= 0.
    void init_potentials(int s) {
        fill(all(pot), LINF);
        pot[s] = 0;
        for (int i = 0; i < n; i++)
            for (int u = 0; u <= n; u++) if (pot[u] !=
                LINF)
                for (auto& e : adj[u]) if (e.cap - e.flow >
                    0 && pot[e.to] > pot[u] + e.cost) pot[e.to] =
                    pot[u] + e.cost;
    }
    bool dijkstra(int s, int t) {
        fill(all(dist), LINF);
        dist[s] = 0;
        priority_queue<pair<ll, int>, vector<pair<ll,
            int>>, greater<>> pq;
        pq.push({0, s});
        while (!pq.empty()) {
            auto [d, u] = pq.top(); pq.pop();
            if (d > dist[u]) continue;
            for (int i = 0; i < adj[u].size(); i++) {
                auto& e = adj[u][i];
                ll rc = e.cost + pot[u] - pot[e.to]; //
                costo reducido
                if (e.cap - e.flow > 0 && dist[e.to] > dist
                    [u] + rc) {
                    dist[e.to] = dist[u] + rc;
                    p_node[e.to] = u; p_edge[e.to] = i;
                    pq.push({dist[e.to], e.to});
                }
            }
        }
        return dist[t] != LINF;
    }
    pair<ll, ll> solve(int s, int t) { // {flujo,
        costo}
        init_potentials(s);
        ll flow = 0, cost = 0;
        while (dijkstra(s, t)) {
            for (int i = 0; i <= n; i++) if (dist[i] !=
                LINF) pot[i] += dist[i];
            ll push = LINF;
            for (int cur = t; cur != s; cur = p_node[cur
                ])

```

```

        push = min(push, adj[p_node[cur]][p_edge[
cur]].cap - adj[p_node[cur]][p_edge[cur]].flow);
    }
    for (int cur = t; cur != s; cur = p_node[cur]) {
        auto& e = adj[p_node[cur]][p_edge[cur]];
        e.flow += push;
        adj[cur][e.rev].flow -= push;
        cost += push * e.cost;
    }
    flow += push;
}
return {flow, cost};
}
};

/**
 * Uso[1]: DominatorTree dt(n); dt.add_edge(u, v);
 * dt.build(root);
 * Arbol dominador de un grafo dirigido desde una
 * raiz; dom[u] son los hijos de u en el arbol.
 * Complejidad:  $O((V + E) \log V)$ .
 */
struct DominatorTree {
    int n, t;
    vector<vi> adj, rev, dom;
    vi dfn, id, p, sd, idom, dsu, best;
    DominatorTree(int n) : n(n), t(0), adj(n + 1),
        rev(n + 1), dom(n + 1),
        dfn(n + 1, 0), id(n + 1), p(n + 1), sd(n + 1),
        idom(n + 1), dsu(n + 1), best(n + 1) {}
    void add_edge(int u, int v) { adj[u].pb(v); }
    void dfs(int u) {
        dfn[u] = ++t;
        id[t] = u;
        sd[t] = best[t] = dsu[t] = t;
        for (int v : adj[u]) {
            if (!dfn[v]) dfs(v), p[dfn[v]] = dfn[u];
            rev[dfn[v]].pb(dfn[u]);
        }
    }
    int find(int x) {
        if (x == dsu[x]) return x;
        int y = find(dsu[x]);
        if (sd[best[x]] > sd[best[dsu[x]]]) best[x] = best[dsu[x]];
        return dsu[x] = y;
    }
    void build(int root) {
        dfs(root);
        vector<vi> bucket(n + 1);
        for (int i = t; i >= 2; i--) {
            for (int u : rev[i]) {
                find(u);
                if (sd[best[u]] < sd[i]) sd[i] = sd[best[u]];
            }
            bucket[sd[i]].pb(i);
            dsu[i] = p[i];
            for (int u : bucket[p[i]]) {
                find(u);
                idom[u] = sd[best[u]] == sd[u] ? sd[u] : best[u];
            }
            bucket[p[i]].clear();
        }
    }
};

```

```

    }
    for (int i = 2; i <= t; i++) {
        if (idom[i] != sd[idom[i]]) idom[i] = idom[idom[i]];
        dom[id[idom[i]]].pb(id[i]);
    }
}
};

```

Trees

```

/**
 * Uso[1]: TreeEdge<T> e = {to, id, weight};
 * Arista de arbol con destino, identificador
 * opcional y peso generico.
 * Complejidad:  $O(1)$  por arista.
 */
template <typename T = ll> struct TreeEdge { int to
    , id; T weight; };

/**
 * Uso[1]: Tree<ll> t(n); t.add_edge(u, v, w); t.
 * init(root); t.get_diameter(); t.get_centers();
 * t.all_farthest_distances();
 * Arbol con raiz: tiempos de entrada/salida (euler
 * ), tamanos de subarbol, diametro con camino y
 * centros.
 * Complejidad: init y diametro  $O(N)$ .
 */
template <typename T = ll> struct Tree {
    int n, root, timer;
    vector<vector<TreeEdge<T>>> adj;
    vi parent, depth, sz, tin, tout, euler_order;
    vector<T> dist_from_root;
    Tree(int n = 0) : n(n), root(1), timer(0), adj(n
        + 1), parent(n + 1), depth(n + 1),
        sz(n + 1), tin(n + 1), tout(n + 1),
        dist_from_root(n + 1) {}
    void add_edge(int u, int v, T w = 1, int id = -1)
    {
        adj[u].pb({v, id, w});
        adj[v].pb({u, id, w});
    }
    void add_directed_edge(int u, int v, T w = 1, int
        id = -1) { adj[u].pb({v, id, w}); }
    void dfs_init(int u, int p = 0, int d = 0, T cur
        = 0) {
        parent[u] = p; depth[u] = d; dist_from_root[u]
        = cur;
        sz[u] = 1; tin[u] = ++timer;
        euler_order.pb(u);
        for (auto& e : adj[u]) if (e.to != p) {
            dfs_init(e.to, u, d + 1, cur + e.weight);
            sz[u] += sz[e.to];
        }
        tout[u] = timer;
    }
    void init(int root = 1) {
        this->root = root; timer = 0;
        euler_order.clear();
        dfs_init(root, 0, 0, 0);
    }
    struct DiameterResult { T length; int u, v; vi
        path; };
    DiameterResult get_diameter() {
        auto farthest = [&](int start) {
            vector<T> dist(n + 1, -1);

```

```

    queue<int> q;
    q.push(start); dist[start] = 0;
    int far = start;
    while (!q.empty()) {
        int u = q.front(); q.pop();
        if (dist[u] > dist[far]) far = u;
        for (auto& e : adj[u]) if (dist[e.to] ==
-1) dist[e.to] = dist[u] + e.weight, q.push(e.
to);
    }
    return pair<int, T>(far, dist[far]);
};

int u = farthest(1).fi;
auto [v, length] = farthest(u);
vi prev(n + 1, -1);
queue<int> q;
q.push(u); prev[u] = 0;
while (!q.empty()) {
    int cur = q.front(); q.pop();
    if (cur == v) break;
    for (auto& e : adj[cur]) if (prev[e.to] ==
-1) prev[e.to] = cur, q.push(e.to);
}
vi path;
for (int cur = v; cur != 0; cur = prev[cur])
    path.pb(cur);
reverse(all(path));
return {length, u, v, path};
}

// 1 o 2 nodos centrales del diametro
vi get_centers() {
    auto d = get_diameter();
    int m = d.path.size();
    if (m & 1) return {d.path[m / 2]};
    return {d.path[m / 2 - 1], d.path[m / 2]};
}

// Distancia maxima desde cada nodo a cualquier
otro. O(N)
vector<T> all_farthest_distances() {
    auto d = get_diameter();
    auto dist_from = [&](int start) {
        vector<T> dist(n + 1, -1);
        queue<int> q;
        q.push(start); dist[start] = 0;
        while (!q.empty()) {
            int cur = q.front(); q.pop();
            for (auto& e : adj[cur]) if (dist[e.to] ==
-1) dist[e.to] = dist[cur] + e.weight, q.push(e.
to);
        }
        return dist;
    };
    auto du = dist_from(d.u), dv = dist_from(d.v);
    vector<T> res(n + 1);
    for (int i = 1; i <= n; i++) res[i] = max(du[i]
, dv[i]);
    return res;
}

/**
 * Uso[1]: BinaryLifting<ll> bl(t, root); bl.lca(u,
v); bl.kth_ancestor(u, k); bl.path_sum(u, v);
bl.dist_weighted(u, v);
 * Binary lifting sobre un arbol: lca, k-esimo

```

```

    ancestro, suma/max/min de aristas en el camino
y distancias.
 * Complejidad: preproceso O(N log N), consulta
O(log N).
 */
template <typename T = ll> struct BinaryLifting {
    int n, log_n;
    vi depth;
    vector<vi> up;
    vector<vector<T>> mx, mn, sum;
    vector<T> dist_root;
    BinaryLifting(const Tree<T>& tree, int root = 1)
    {
        n = tree.n;
        log_n = 32 - __builtin_clz(n) + 1;
        depth = tree.depth;
        dist_root = tree.dist_from_root;
        up.assign(n + 1, vi(log_n, 0));
        mx.assign(n + 1, vector<T>(log_n, -LINF));
        mn.assign(n + 1, vector<T>(log_n, LINF));
        sum.assign(n + 1, vector<T>(log_n, 0));
        for (int u = 1; u <= n; u++) up[u][0] = tree.
parent[u] ? tree.parent[u] : u;
        for (int u = 1; u <= n; u++)
            for (auto& e : tree.adj[u]) if (e.to == tree.
parent[u]) mx[u][0] = mn[u][0] = sum[u][0] = e.
weight;
        for (int j = 1; j < log_n; j++)
            for (int i = 1; i <= n; i++) {
                int p = up[i][j - 1];
                up[i][j] = up[p][j - 1];
                mx[i][j] = max(mx[i][j - 1], mx[p][j - 1]);
                mn[i][j] = min(mn[i][j - 1], mn[p][j - 1]);
                sum[i][j] = sum[i][j - 1] + sum[p][j - 1];
            }
    }

    int kth_ancestor(int u, int k) {
        if (depth[u] < k) return 0;
        for (int j = 0; j < log_n; j++) if (k & (1 << j
)) u = up[u][j];
        return u;
    }

    int lca(int u, int v) {
        if (depth[u] < depth[v]) swap(u, v);
        u = kth_ancestor(u, depth[u] - depth[v]);
        if (u == v) return u;
        for (int j = log_n - 1; j >= 0; j--) if (up[u][
j] != up[v][j]) u = up[u][j], v = up[v][j];
        return up[u][0];
    }

    int dist_edges(int u, int v) { return depth[u] +
depth[v] - 2 * depth[lca(u, v)]; }
    T dist_weighted(int u, int v) { return dist_root[
u] + dist_root[v] - 2 * dist_root[lca(u, v)]; }
    // k-esimo nodo en el camino u->v (0-indexed: 0 =
u)
    int kth_node_on_path(int u, int v, int k) {
        int anc = lca(u, v);
        int d1 = depth[u] - depth[anc], d2 = depth[v] -
depth[anc];
        if (k <= d1) return kth_ancestor(u, k);
        k -= d1;
        if (k <= d2) return kth_ancestor(v, d2 - k);
        return 0;
    }

    T path_max(int u, int v) {

```

```

    int anc = lca(u, v);
    T res = -LINF;
    auto lift = [&](int node, int steps) {
        for (int j = 0; j < log_n; j++) if (steps &
(1 << j)) res = max(res, mx[node][j]), node =
up[node][j];
    };
    lift(u, depth[u] - depth[anc]);
    lift(v, depth[v] - depth[anc]);
    return res;
}
T path_min(int u, int v) {
    int anc = lca(u, v);
    T res = LINF;
    auto lift = [&](int node, int steps) {
        for (int j = 0; j < log_n; j++) if (steps &
(1 << j)) res = min(res, mn[node][j]), node =
up[node][j];
    };
    lift(u, depth[u] - depth[anc]);
    lift(v, depth[v] - depth[anc]);
    return res;
}
T path_sum(int u, int v) {
    int anc = lca(u, v);
    T res = 0;
    auto lift = [&](int node, int steps) {
        for (int j = 0; j < log_n; j++) if (steps &
(1 << j)) res += sum[node][j], node = up[node][
j];
    };
    lift(u, depth[u] - depth[anc]);
    lift(v, depth[v] - depth[anc]);
    return res;
}
bool is_ancestor(int u, int v) { return lca(u, v)
== u; }
};

/**
 * Uso[1]: EulerTourTree<ll> et(t); et.
 * subtree_range(u); et.subtree_size(u);
 * Mapea el subarbol de cada nodo a un rango
 * contiguo de la secuencia de entrada (tin/tout).
 * Complejidad: O(1) por consulta.
 */
template <typename T = ll> struct EulerTourTree {
    vi in, out;
    EulerTourTree(const Tree<T>& tree) : in(tree.tin)
, out(tree.tout) {}
    pair<int, int> subtree_range(int u) { return {in[
u], out[u]}; }
    int subtree_size(int u) { return out[u] - in[u] +
1; }
};

/**
 * Uso[1]: HLD<ll> hld(n); hld.add_edge(u, v); hld.
 * init(root); hld.update_path(u, v, val); hld.
 * query_path(u, v);
 * Heavy-light decomposition con segment tree
 * perezoso para sumas y updates sobre caminos y
 * subarboles.
 * Complejidad: preproceso O(N), update y query de
 * camino O(log2 N).
 */

```

```

template <typename T = ll> struct HLD {
    struct SegTree {
        int size;
        vector<T> tree, lazy;
        void init(int n) {
            size = 1;
            while (size < n) size <= 1;
            tree.assign(2 * size, 0);
            lazy.assign(2 * size, 0);
        }
        void push(int node, int l, int r) {
            if (!lazy[node]) return;
            tree[node] += lazy[node] * (r - l + 1);
            if (l != r) lazy[2 * node] += lazy[node],
lazy[2 * node + 1] += lazy[node];
            lazy[node] = 0;
        }
        void update(int node, int l, int r, int ql, int
qr, T val) {
            push(node, l, r);
            if (ql > r || qr < l) return;
            if (ql <= l && r <= qr) { lazy[node] += val;
push(node, l, r); return; }
            int mid = (l + r) / 2;
            update(2 * node, l, mid, ql, qr, val);
            update(2 * node + 1, mid + 1, r, ql, qr, val)
;
            tree[node] = tree[2 * node] + tree[2 * node +
1];
        }
        T query(int node, int l, int r, int ql, int qr)
        {
            push(node, l, r);
            if (ql > r || qr < l) return 0;
            if (ql <= l && r <= qr) return tree[node];
            int mid = (l + r) / 2;
            return query(2 * node, l, mid, ql, qr) +
query(2 * node + 1, mid + 1, r, ql, qr);
        }
    } seg;
    int n, cur_pos;
    vector<vi> adj;
    vi parent, depth, heavy, head, pos;
    HLD(int n) : n(n), cur_pos(1), adj(n + 1), parent
(n + 1), depth(n + 1),
heavy(n + 1, -1), head(n + 1), pos(n + 1) { seg
.init(n + 1); }
    void add_edge(int u, int v) { adj[u].pb(v); adj[v]
.pb(u); }
    int dfs_size(int u, int p, int d) {
        int sz = 1, max_c = 0;
        depth[u] = d; parent[u] = p; heavy[u] = -1;
        for (int v : adj[u]) if (v != p) {
            int cs = dfs_size(v, u, d + 1);
            sz += cs;
            if (cs > max_c) max_c = cs, heavy[u] = v;
        }
        return sz;
    }
    void decompose(int u, int h) {
        head[u] = h;
        pos[u] = cur_pos++;
        if (heavy[u] != -1) decompose(heavy[u], h);
        for (int v : adj[u]) if (v != parent[u] && v !=
heavy[u]) decompose(v, v);
    }
}

```

```

}
void init(int root = 1) { cur_pos = 1; dfs_size(
    root, 0, 0); decompose(root, root); }
void update_path(int u, int v, T val) {
    while (head[u] != head[v]) {
        if (depth[head[u]] > depth[head[v]]) swap(u,
            v);
        seg.update(1, 1, n, pos[head[v]], pos[v], val);
        v = parent[head[v]];
    }
    if (depth[u] > depth[v]) swap(u, v);
    seg.update(1, 1, n, pos[u], pos[v], val);
}
T query_path(int u, int v) {
    T res = 0;
    while (head[u] != head[v]) {
        if (depth[head[u]] > depth[head[v]]) swap(u,
            v);
        res += seg.query(1, 1, n, pos[head[v]], pos[v]);
        v = parent[head[v]];
    }
    if (depth[u] > depth[v]) swap(u, v);
    return res + seg.query(1, 1, n, pos[u], pos[v]);
}
void update_subtree(int u, int sz_u, T val) { seg
    .update(1, 1, n, pos[u], pos[u] + sz_u - 1, val); }
T query_subtree(int u, int sz_u) { return seg.
    query(1, 1, n, pos[u], pos[u] + sz_u - 1); }
void update_node(int u, T val) { seg.update(1, 1,
    n, pos[u], pos[u], val); }
int lca(int u, int v) {
    while (head[u] != head[v]) {
        if (depth[head[u]] > depth[head[v]]) swap(u,
            v);
        v = parent[head[v]];
    }
    return depth[u] < depth[v] ? u : v;
}
};

/**
 * Uso[1]: CentroidDecomposition cd(n); cd.add_edge
 * (u, v, w); cd.K = lim; cd.init(true); cd.
 * total_paths;
 * Descomposicion en centroides: conteo offline de
 * pares con distancia menor o igual a K y
 * consultas online de nodo mas cercano.
 * Complejidad: construccion  $O(N \log N)$ , update y
 * query online  $O(\log N)$ .
 */
struct CentroidDecomposition {
    int n;
    vector<vector<pair<int, ll>>> adj;
    vi sz, c_parent;
    vector<bool> removed;
    ll total_paths = 0, K = 0;
    vll local_ans;
    CentroidDecomposition(int n) : n(n), adj(n + 1),
        sz(n + 1), c_parent(n + 1),
        removed(n + 1, false), local_ans(n + 1, LINF)
    {}
    void add_edge(int u, int v, ll w = 1) { adj[u].pb
        ({v, w}); adj[v].pb({u, w}); }
    int get_sizes(int u, int p = 0) {
        sz[u] = 1;
        for (auto [v, w] : adj[u]) if (v != p && !
            removed[v]) sz[u] += get_sizes(v, u);
        return sz[u];
    }
    int get_centroid(int u, int p, int size) {
        for (auto [v, w] : adj[u]) if (v != p && !
            removed[v] && sz[v] > size / 2) return
            get_centroid(v, u, size);
        return u;
    }
    void get_paths(int u, int p, ll dist, vll& paths)
    {
        paths.pb(dist);
        for (auto [v, w] : adj[u]) if (v != p && !
            removed[v]) get_paths(v, u, dist + w, paths);
    }
    ll count_pairs(vll& paths, ll limit) {
        sort(all(paths));
        ll cnt = 0;
        for (int l = 0, r = paths.size() - 1; l < r;) {
            if (paths[l] + paths[r] <= limit) cnt += r -
                l, l++;
            else r--;
        }
        return cnt;
    }
    void process_centroid(int c) {
        vll all_paths{0};
        for (auto [v, w] : adj[c]) if (!removed[v])
            get_paths(v, c, w, all_paths);
        total_paths += count_pairs(all_paths, K);
        for (auto [v, w] : adj[c]) if (!removed[v]) {
            vll sub;
            get_paths(v, c, w, sub);
            total_paths -= count_pairs(sub, K);
        }
    }
    int build_tree(int u, int p = 0, bool
        offline_mode = false) {
        int c = get_centroid(u, 0, get_sizes(u, 0));
        if (offline_mode) process_centroid(c);
        removed[c] = true;
        c_parent[c] = p;
        for (auto [v, w] : adj[c]) if (!removed[v])
            build_tree(v, c, offline_mode);
        return c;
    }
    void init(bool offline_mode = false) { build_tree
        (1, 0, offline_mode); }
    // Pinta u de rojo. LCA del arbol original.
    void update(int u, LCA& lca) {
        for (int cur = u; cur; cur = c_parent[cur])
            local_ans[cur] = min(local_ans[cur], (ll)lca.
                dist(u, cur));
    }
    ll query(int u, LCA& lca) {
        ll best = LINF;
        for (int cur = u; cur; cur = c_parent[cur])
            best = min(best, (ll)lca.dist(u, cur) +
                local_ans[cur]);
        return best;
    }
}

```

```

};

/**
 * Uso[1]: DSUOnTree s(n); s.add_edge(u, v); s.
set_colors(color_lindex); s.solve(root); s.ans[
u];
 * Small-to-large (sack) para responder
estadísticas de colores en cada subarbol.
 * Complejidad:  $O(N \log N)$ .
 */
struct DSUOnTree {
    int n, timer, distinct;
    vector<vi> adj;
    vi sz, heavy, tin, tout, euler_node, color, ans,
    cnt;
    DSUOnTree(int n) : n(n), timer(0), distinct(0),
        adj(n + 1), sz(n + 1), heavy(n + 1, -1),
        tin(n + 1), tout(n + 1), euler_node(n + 1),
        color(n + 1), ans(n + 1), cnt(n + 1) {}
    void add_edge(int u, int v) { adj[u].pb(v); adj[v]
        .pb(u); }
    void set_colors(const vi& c) {
        for (int i = 1; i <= n; i++) color[i] = c[i];
    }
    void dfs_size(int u, int p = 0) {
        sz[u] = 1;
        tin[u] = ++timer;
        euler_node[timer] = u;
        int max_sz = 0;
        for (int v : adj[u]) if (v != p) {
            dfs_size(v, u);
            sz[u] += sz[v];
            if (sz[v] > max_sz) max_sz = sz[v], heavy[u]
            = v;
        }
        tout[u] = timer;
    }
    void add_node(int u) { if (!cnt[color[u]]++)
        distinct++; }
    void remove_node(int u) { if (--cnt[color[u]])
        distinct--; }
    void dfs_sack(int u, int p, bool keep) {
        for (int v : adj[u]) if (v != p && v != heavy[u])
            dfs_sack(v, u, false);
        if (heavy[u] != -1) dfs_sack(heavy[u], u, true);
        add_node(u);
        for (int v : adj[u]) if (v != p && v != heavy[u])
            for (int t = tin[v]; t <= tout[v]; t++)
                add_node(euler_node[t]);
        ans[u] = distinct;
        if (!keep) for (int t = tin[u]; t <= tout[u]; t
            ++) remove_node(euler_node[t]);
    }
    void solve(int root = 1) { timer = 0; dfs_size(
        root); dfs_sack(root, 0, false); }
};

/**
 * Uso[1]: VirtualTree vt(t, bl); auto [raiz, nodos
] = vt.build(vector_de_nodos);
 * Construye el arbol virtual sobre un subconjunto
de nodos, con aristas pesadas por distancia.
 * Complejidad:  $O(K \log N)$  con  $K$  el numero de nodos
solicitados.

```

```

 */
struct VirtualTree {
    const Tree<ll>& tree;
    BinaryLifting<ll>& bl;
    vector<vector<pair<int, ll>>> virt_adj;
    VirtualTree(const Tree<ll>& tree, BinaryLifting<
        ll>& bl) : tree(tree), bl(bl), virt_adj(tree.n
        + 1) {}
    // Retorna {raiz, nodos del arbol virtual}
    pair<int, vi> build(vi nodes) {
        sort(all(nodes), [&](int a, int b) { return
            tree.tin[a] < tree.tin[b]; });
        int k = nodes.size();
        for (int i = 0; i + 1 < k; i++) nodes.pb(bl.lca
            (nodes[i], nodes[i + 1]));
        sort(all(nodes), [&](int a, int b) { return
            tree.tin[a] < tree.tin[b]; });
        nodes.erase(unique(all(nodes)), nodes.end());
        for (int u : nodes) virt_adj[u].clear();
        vi stk{nodes[0]};
        for (int i = 1; i < nodes.size(); i++) {
            int u = nodes[i];
            while (!stk.empty() && !bl.is_ancestor(stk.
                back(), u)) stk.pop_back();
            if (!stk.empty()) {
                ll w = bl.dist_weighted(stk.back(), u);
                virt_adj[stk.back()].pb({u, w});
                virt_adj[u].pb({stk.back(), w});
            }
            stk.pb(u);
        }
        return {nodes[0], nodes};
    }
};

/**
 * Uso[1]: TreeHashing::are_isomorphic(n1, adj1, n2
, adj2);
 * Determina si dos arboles son isomorfos (sin
considerar etiquetas) mediante hashing y
centroides.
 * Complejidad:  $O(N)$ .
 */
struct TreeHashing {
    static uint64_t splitmix64(uint64_t x) {
        x += 0x9e3779b97f4a7c15;
        x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
        x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
        return x ^ (x >> 31);
    }
    static uint64_t hash_node(uint64_t x) { return
        splitmix64(x + 0x517cc1b727220a95); }
    static uint64_t rooted_hash(int u, int p, const
        vector<vi>& adj) {
        vector<uint64_t> child;
        for (int v : adj[u]) if (v != p) child.pb(
            rooted_hash(v, u, adj));
        sort(all(child));
        uint64_t h = 1;
        for (uint64_t c : child) h += hash_node(c);
        return splitmix64(h);
    }
    static vector<int> get_centers(int n, const
        vector<vi>& adj) {
        vi deg(n + 1);

```

```

queue<int> leaves;
for (int i = 1; i <= n; i++) {
    deg[i] = adj[i].size();
    if (deg[i] <= 1) leaves.push(i);
}
int remaining = n;
while (remaining > 2) {
    int sz = leaves.size();
    remaining -= sz;
    for (int i = 0; i < sz; i++) {
        int u = leaves.front(); leaves.pop();
        for (int v : adj[u]) if (--deg[v] == 1)
            leaves.push(v);
    }
}
vi centers;
while (!leaves.empty()) centers.pb(leaves.front()), leaves.pop();
return centers;
}

static bool are_isomorphic(int n1, const vector<
vi>& adj1, int n2, const vector<vi>& adj2) {
    if (n1 != n2) return false;
    auto c1 = get_centers(n1, adj1), c2 =
get_centers(n2, adj2);
    for (int r1 : c1)
        for (int r2 : c2)
            if (rooted_hash(r1, 0, adj1) == rooted_hash(
r2, 0, adj2)) return true;
    return false;
}
};

/**
 * Uso[1]: RerootingDP r(n); r.add_edge(u, v); auto
ans = r.solve(root);
 * DP con rerooting para calcular la solucion para
todas las posibles raices; ajustar merge,
extend y finalize.
 * Complejidad: O(N).
 */
struct RerootingDP {
    struct State { ll val; };
    int n;
    vector<vi> adj;
    vector<State> dp_down, ans;
    const State NEUTRAL = {0};
    State merge(State a, State b) { return {max(a.val
, b.val)}; }
    State extend(State s, int v, int u) { return {s.
val + 1}; } // transicion por arista
    State finalize(State s, int u) { return s; }
    RerootingDP(int n) : n(n), adj(n + 1), dp_down(n
+ 1, NEUTRAL), ans(n + 1, NEUTRAL) {}
    void add_edge(int u, int v) { adj[u].pb(v); adj[v
].pb(u); }
    void dfs_down(int u, int p) {
        State cur = NEUTRAL;
        for (int v : adj[u]) if (v != p) dfs_down(v, u)
, cur = merge(cur, extend(dp_down[v], v, u));
        dp_down[u] = finalize(cur, u);
    }
    void dfs_reroot(int u, int p, State from_parent)
    {
        vi children;
        for (int v : adj[u]) if (v != p) children.pb(v)
;
        int k = children.size();
        vector<State> pref(k + 1, NEUTRAL), suff(k + 1,
NEUTRAL);
        for (int i = 0; i < k; i++) pref[i + 1] = merge
(pref[i], extend(dp_down[children[i]], children
[i], u));
        for (int i = k - 1; i >= 0; i--) suff[i] =
merge(suff[i + 1], extend(dp_down[children[i]],
children[i], u));
        ans[u] = finalize(merge(pref[k], from_parent),
u);
        for (int i = 0; i < k; i++) {
            State outside = merge(merge(pref[i], suff[i +
1]), from_parent);
            dfs_reroot(children[i], u, extend(finalize(
outside, u), u, children[i]));
        }
    }
    vector<State> solve(int root = 1) {
        dfs_down(root, 0);
        dfs_reroot(root, 0, NEUTRAL);
        return ans;
    }
};

/**
 * Uso[1]: FastLCA<ll> fl(t); fl.lca(u, v); fl.
dist_edges(u, v);
 * LCA por Euler + RMQ con respuesta en tiempo
constante por consulta.
 * Complejidad: preproceso O(N log N), consulta O(1).
 */
template <typename T = ll> struct FastLCA {
    const Tree<T>& tree;
    vi tin, euler, euler_depth, log_table;
    vector<vi> st;
    FastLCA(const Tree<T>& tree) : tree(tree), tin(
tree.n + 1) {
        auto dfs = [&](auto&& self, int u, int p, int d
) -> void {
            tin[u] = euler.size();
            euler.pb(u);
            euler_depth.pb(d);
            for (auto& e : tree.adj[u]) if (e.to != p) {
                self(self, e.to, u, d + 1);
                euler.pb(u);
                euler_depth.pb(d);
            }
        };
        dfs(dfs, tree.root, 0, 0);
        int m = euler.size();
        log_table.assign(m + 1, 0);
        for (int i = 2; i <= m; i++) log_table[i] =
log_table[i / 2] + 1;
        int K = log_table[m] + 1;
        st.assign(K, vi(m));
        for (int i = 0; i < m; i++) st[0][i] = i;
        for (int i = 1; i < K; i++)
            for (int j = 0; j + (1 << i) <= m; j++) {
                int l = st[i - 1][j], r = st[i - 1][j + (1
<< (i - 1))];
                st[i][j] = euler_depth[l] < euler_depth[r]
? l : r;
            }
    }
};

```

```

}
int lca(int u, int v) {
    int l = tin[u], r = tin[v];
    if (l > r) swap(l, r);
    int i = log_table[r - 1 + 1];
    int a = st[i][l], b = st[i][r - (1 << i) + 1];
    return euler_depth[a] < euler_depth[b] ? euler[
        a] : euler[b];
}
int dist_edges(int u, int v) { return tree.depth[
    u] + tree.depth[v] - 2 * tree.depth[lca(u, v)];
}
};

/**
 * Uso[1]: MoOnTrees<ll> mo(n); mo.dfs_euler(root,
 * 0, t); mo.solve(queries, fast_lca);
 * Algoritmo de Mo sobre caminos de un arbol;
 * ajustar toggle para actualizar la respuesta al
 * agregar o quitar un nodo.
 * Complejidad:  $O((N + Q)\sqrt{N})$  toggle esperados.
 */
template <typename T = ll> struct MoOnTrees {
    struct Query {
        int l, r, lca, id, block_size;
        bool operator<(const Query& o) const {
            int b1 = l / block_size, b2 = o.l /
                block_size;
            if (b1 != b2) return b1 < b2;
            return (b1 & 1) ? (r < o.r) : (r > o.r);
        }
    };
    int n, timer;
    vi tin, tout, euler;
    vector<bool> in_path;
    ll current_ans = 0;
    MoOnTrees(int n) : n(n), timer(0), tin(n + 1),
        tout(n + 1), euler(2 * n + 1), in_path(n + 1,
            false) {}
    void dfs_euler(int u, int p, const Tree<T>& tree)
    {
        tin[u] = ++timer;
        euler[timer] = u;
        for (auto& e : tree.adj[u]) if (e.to != p)
            dfs_euler(e.to, u, tree);
        tout[u] = ++timer;
        euler[timer] = u;
    }
    void toggle(int u) {
        // TODO: actualizar respuesta al agregar/
        // remover el nodo u (p.ej. frecuencias de colores)
        in_path[u] = !in_path[u];
    }
    vll solve(vector<pair<int, int>>& raw_queries,
        const FastLCA<T>& fast_lca) {
        int q = raw_queries.size();
        int bs = max(1, (int)(2 * n / sqrt(max(1, q))))
            ;
        vector<Query> queries;
        for (int i = 0; i < q; i++) {
            auto [u, v] = raw_queries[i];
            if (tin[u] > tin[v]) swap(u, v);
            int anc = fast_lca.lca(u, v);
            queries.pb(anc == u ? Query{tin[u], tin[v],
                0, i, bs} : Query{tout[u], tin[v], anc, i, bs})
        }

```

```

};
sort(all(queries));
vll ans(q);
int curr_l = 1, curr_r = 0;
for (auto& mq : queries) {
    while (curr_l > mq.l) toggle(euler[--curr_l]);
    while (curr_r < mq.r) toggle(euler[++curr_r]);
    while (curr_l < mq.l) toggle(euler[curr_l++]);
    while (curr_r > mq.r) toggle(euler[curr_r--]);
    if (mq.lca) toggle(mq.lca);
    ans[mq.id] = current_ans;
    if (mq.lca) toggle(mq.lca);
}
return ans;
};

```

Range Queries

```

/**
 * Uso[0]: SparseTable st(a); st.query(L, R);
 * Consultas estaticas en rangos para operaciones
 * idempotentes (min, max, gcd).
 * Complejidad: construccion  $O(N \log N)$ , consulta
 *  $O(1)$ .
 */
struct SparseTable {
    int n;
    vector<vll> st;
    SparseTable(const vll& a) : n(a.size()) {
        int log_n = log2(n) + 1;
        st.assign(n, vll(log_n));
        for (int i = 0; i < n; i++) st[i][0] = a[i];
        for (int j = 1; j < log_n; j++)
            for (int i = 0; i + (1 << j) <= n; i++)
                st[i][j] = min(st[i][j - 1], st[i + (1 <<
                    j - 1)][j - 1]); // min/max/gcd
    }
    ll query(int L, int R) {
        int j = log2(R - L + 1);
        return min(st[L][j], st[R - (1 << j) + 1][j]);
    }
};

/**
 * Uso[1]: FenwickTree ft(n); ft.add(i, x); ll s =
 * ft.query(2, 3);
 * Arbol de Fenwick para sumas de prefijos y rangos
 * con actualizaciones puntuales.
 * Complejidad: actualizacion y consulta  $O(\log N)$ .
 */
struct FenwickTree {
    int n;
    vll bit;
    FenwickTree(int n) : n(n), bit(n + 1, 0) {}
    void add(int idx, ll val) { for (; idx <= n; idx
        += idx & -idx) bit[idx] += val; }
    ll sum(int idx) {
        ll res = 0;
        for (; idx > 0; idx -= idx & -idx) res += bit[
            idx];
        return res;
    }
}

```

```

11 query(int L, int R) { return sum(R) - sum(L -
    1); }
};

/**
 * Uso[0]: IterativeSegTree st(a); st.update(pos,
    val); st.query(L, R);
 * Arbol de segmentos iterativo para suma (o min/
    max) con actualizacion puntual.
 * Complejidad: actualizacion y consulta O(log N).
 */
struct IterativeSegTree {
    int n;
    vll st;
    IterativeSegTree(const vll& a) : n(a.size()), st
        (2 * a.size(), 0) {
        for (int i = 0; i < n; i++) st[n + i] = a[i];
        for (int i = n - 1; i > 0; i--) st[i] = st[i <<
            1] + st[i << 1 | 1]; // + / min / max
    }
    void update(int pos, ll val) {
        for (st[pos += n] = val; pos > 1; pos >>= 1) st
            [pos >> 1] = st[pos] + st[pos ^ 1];
    }
    ll query(int L, int R) {
        ll resL = 0, resR = 0;
        for (L += n, R += n + 1; L < R; L >>= 1, R >>=
            1) {
            if (L & 1) resL += st[L++];
            if (R & 1) resR += st[--R];
        }
        return resL + resR;
    }
};

/**
 * Uso[0]: RecursiveSegTree st(a); st.update(pos,
    val); st.query(L, R); st.find_first(val);
 * Arbol de segmentos recursivo para max (o suma/
    min) con busqueda del primer indice con valor
    mayor o igual.
 * Complejidad: actualizacion y consulta O(log N).
 */
struct RecursiveSegTree {
    int n;
    vll st;
    ll combine(ll a, ll b) { return max(a, b); } //
        max / + / min
    RecursiveSegTree(const vll& a) : n(a.size()), st
        (4 * a.size()) { build(1, 0, n - 1, a); }
    void build(int p, int L, int R, const vll& a) {
        if (L == R) { st[p] = a[L]; return; }
        int mid = (L + R) / 2;
        build(p << 1, L, mid, a);
        build(p << 1 | 1, mid + 1, R, a);
        st[p] = combine(st[p << 1], st[p << 1 | 1]);
    }
    void update(int p, int L, int R, int pos, ll val)
        {
        if (L == R) { st[p] = val; return; } // o st[p]
            += val
        int mid = (L + R) / 2;
        pos <= mid ? update(p << 1, L, mid, pos, val) :
            update(p << 1 | 1, mid + 1, R, pos, val);
        st[p] = combine(st[p << 1], st[p << 1 | 1]);
    }
};

```

```

11 query(int p, int L, int R, int qL, int qR) {
    if (qL > R || qR < L) return 0; // NEUTRAL
    if (qL <= L && R <= qR) return st[p];
    int mid = (L + R) / 2;
    return combine(query(p << 1, L, mid, qL, qR),
        query(p << 1 | 1, mid + 1, R, qL, qR));
}
// Primer indice con valor >= val (asume arbol de
    max). -1 si no existe
int find_first(int p, int L, int R, ll val) {
    if (st[p] < val) return -1;
    if (L == R) return L;
    int mid = (L + R) / 2;
    return st[p << 1] >= val ? find_first(p << 1, L,
        mid, val) : find_first(p << 1 | 1, mid + 1, R,
        val);
}
void update(int pos, ll val) { update(1, 0, n -
    1, pos, val); }
ll query(int L, int R) { return query(1, 0, n -
    1, L, R); }
int find_first(ll val) { return find_first(1, 0,
    n - 1, val); }
};

/**
 * Uso[0]: LazySegTree st(a); st.add_range(L, R, v)
    ; st.set_range(L, R, v); st.query(L, R); st.get
    (pos);
 * Arbol de segmentos con propagacion perezosa para
    suma con actualizaciones de rango (add y set).
 * Complejidad: actualizacion y consulta de rango
    O(log N).
 */
struct LazySegTree {
    int n;
    vll st, lazy_add, lazy_set;
    vector<bool> marked;
    LazySegTree(const vll& a) : n(a.size()), st(4 * a
        .size()), lazy_add(4 * a.size()),
        lazy_set(4 * a.size()), marked(4 * a.size()),
        false { build(1, 0, n - 1, a); }
    void apply_set(int p, ll val, int L, int R) {
        st[p] = val * (R - L + 1); // para min/max: st[
            p] = val
        lazy_set[p] = val;
        lazy_add[p] = 0;
        marked[p] = true;
    }
    void apply_add(int p, ll val, int L, int R) {
        st[p] += val * (R - L + 1); // para min/max: st
            [p] += val
        if (marked[p]) lazy_set[p] += val;
        else lazy_add[p] += val;
    }
    void push(int p, int L, int R) {
        int mid = (L + R) / 2;
        if (marked[p]) {
            apply_set(p << 1, lazy_set[p], L, mid);
            apply_set(p << 1 | 1, lazy_set[p], mid + 1, R
            );
            marked[p] = false;
        }
        if (lazy_add[p]) {
            apply_add(p << 1, lazy_add[p], L, mid);

```

```

    apply_add(p << 1 | 1, lazy_add[p], mid + 1, R
);
    lazy_add[p] = 0;
}
}
void build(int p, int L, int R, const vll& a) {
    if (L == R) { st[p] = a[L]; return; }
    int mid = (L + R) / 2;
    build(p << 1, L, mid, a);
    build(p << 1 | 1, mid + 1, R, a);
    st[p] = st[p << 1] + st[p << 1 | 1];
}
void update(int p, int L, int R, int qL, int qR,
    ll val, int type) { // 1 = add, 2 = set
    if (qL > R || qR < L) return;
    if (qL <= L && R <= qR) { type == 1 ? apply_add
        (p, val, L, R) : apply_set(p, val, L, R);
        return; }
    push(p, L, R);
    int mid = (L + R) / 2;
    update(p << 1, L, mid, qL, qR, val, type);
    update(p << 1 | 1, mid + 1, R, qL, qR, val,
        type);
    st[p] = st[p << 1] + st[p << 1 | 1];
}
ll query(int p, int L, int R, int qL, int qR) {
    if (qL > R || qR < L) return 0;
    if (qL <= L && R <= qR) return st[p];
    push(p, L, R);
    int mid = (L + R) / 2;
    return query(p << 1, L, mid, qL, qR) + query(p
        << 1 | 1, mid + 1, R, qL, qR);
}
ll get_point(int p, int L, int R, int pos) {
    if (L == R) return st[p];
    push(p, L, R);
    int mid = (L + R) / 2;
    return pos <= mid ? get_point(p << 1, L, mid,
        pos) : get_point(p << 1 | 1, mid + 1, R, pos);
}
void add_range(int L, int R, ll val) { update(1,
    0, n - 1, L, R, val, 1); }
void set_range(int L, int R, ll val) { update(1,
    0, n - 1, L, R, val, 2); }
ll query(int L, int R) { return query(1, 0, n -
    1, L, R); }
ll get(int pos) { return get_point(1, 0, n - 1,
    pos); }
};

/**
 * Uso[1]: DynamicSegTree st(max_rango); st.update(
    pos, val); st.query(L, R);
 * Arbol de segmentos dinamico (sparse) para
    indices grandes sin compresion previa.
 * Complejidad: actualizacion y consulta
    O(log max_rango).
 */
struct DynamicSegTree {
    struct Node { ll sum = 0; int left = 0, right =
        0; };
    vector<Node> st;
    ll max_range;
    int root;
    DynamicSegTree(ll max_range = 1e9) : max_range(

```

```

        max_range), root(0) { st.emplace_back(); }
    int new_node() { st.emplace_back(); return st.
        size() - 1; }
    int update(int node, ll L, ll R, ll pos, ll val)
    {
        if (!node) node = new_node();
        st[node].sum += val;
        if (L == R) return node;
        ll mid = (L + R) / 2;
        if (pos <= mid) st[node].left = update(st[node]
            .left, L, mid, pos, val);
        else st[node].right = update(st[node].right,
            mid + 1, R, pos, val);
        return node;
    }
    ll query(int node, ll L, ll R, ll qL, ll qR) {
        if (!node || qL > R || qR < L) return 0;
        if (qL <= L && R <= qR) return st[node].sum;
        ll mid = (L + R) / 2;
        return query(st[node].left, L, mid, qL, qR) +
            query(st[node].right, mid + 1, R, qL, qR);
    }
    void update(ll pos, ll val) { root = update(root,
        1, max_range, pos, val); }
    ll query(ll L, ll R) { return query(root, 1,
        max_range, L, R); }
};

/**
 * Uso[0]: PersistentSegTree pst(a); int v = pst.
    update(version, pos, val); pst.query(v, L, R);
 * Arbol de segmentos persistente con versiones
    historicas y consultas sobre cualquier version.
 * Complejidad: actualizacion y consulta O(log N);
    espacio O(N log N).
 */
struct PersistentSegTree {
    struct Node { ll sum; int left, right; Node(ll s
        = 0, int l = 0, int r = 0) : sum(s), left(l),
        right(r) {} };
    vector<Node> st;
    vi roots;
    int n;
    int new_node(ll sum, int left, int right) { st.
        emplace_back(sum, left, right); return st.size
        () - 1; }
    int build(int L, int R, const vll& a) {
        if (L == R) return new_node(a[L], 0, 0);
        int mid = (L + R) / 2;
        int l = build(L, mid, a), r = build(mid + 1, R,
            a);
        return new_node(st[l].sum + st[r].sum, l, r);
    }
    int update(int old, int L, int R, int pos, ll val)
    {
        if (L == R) return new_node(st[old].sum + val,
            0, 0);
        int mid = (L + R) / 2;
        int l = st[old].left, r = st[old].right;
        if (pos <= mid) l = update(l, L, mid, pos, val)
            ;
        else r = update(r, mid + 1, R, pos, val);
        return new_node(st[l].sum + st[r].sum, l, r);
    }
    ll query(int node, int L, int R, int qL, int qR)

```

```

{
    if (!node || qL > R || qR < L) return 0;
    if (qL <= L && R <= qR) return st[node].sum;
    int mid = (L + R) / 2;
    return query(st[node].left, L, mid, qL, qR) +
        query(st[node].right, mid + 1, R, qL, qR);
}
PersistentSegTree(const vll& a) : n(a.size()) {
    st.emplace_back();
    roots.pb(build(0, n - 1, a));
}
int update(int prev_version, int pos, ll val) {
    // crea nueva version
    roots.pb(update(roots[prev_version], 0, n - 1,
        pos, val));
    return roots.size() - 1;
}
ll query(int version, int L, int R) { return
    query(roots[version], 0, n - 1, L, R); }
};

/**
 * Uso[1]: FenwickTree2D ft(n, m); ft.add(r, c, v);
 * ft.query(r1, c1, r2, c2);
 * Arbol de Fenwick bidimensional para sumas de
 * submatrices con actualizaciones puntuales.
 * Complejidad: actualizacion y consulta
 *  $O(\log n \cdot \log m)$ .
 */
struct FenwickTree2D {
    int n, m;
    vector<vll> bit;
    FenwickTree2D(int n, int m) : n(n), m(m), bit(n +
        1, vll(m + 1, 0)) {}
    void add(int r, int c, ll val) {
        for (int i = r; i <= n; i += i & -i)
            for (int j = c; j <= m; j += j & -j) bit[i][j]
                += val;
    }
    ll sum(int r, int c) {
        ll res = 0;
        for (int i = r; i > 0; i -= i & -i)
            for (int j = c; j > 0; j -= j & -j) res +=
                bit[i][j];
        return res;
    }
    ll query(int r1, int c1, int r2, int c2) {
        return sum(r2, c2) - sum(r1 - 1, c2) - sum(r2,
            c1 - 1) + sum(r1 - 1, c1 - 1);
    }
};

/**
 * Uso[0]: MergeSortTree st(a); st.count_less_than(
 * L, R, val); st.count_less_equal(L, R, val);
 * Arbol de segmentos con listas ordenadas para
 * contar elementos menores o iguales a un valor
 * en un rango.
 * Complejidad: construccion  $O(N \log N)$ , consulta
 *  $O(\log^2 N)$ .
 */
struct MergeSortTree {
    int n;
    vector<vll> st;
    MergeSortTree(const vll& a) : n(a.size()), st(4 *
        a.size()) { build(1, 0, n - 1, a); }

```

```

void build(int p, int L, int R, const vll& a) {
    if (L == R) { st[p] = {a[L]}; return; }
    int mid = (L + R) / 2;
    build(p << 1, L, mid, a);
    build(p << 1 | 1, mid + 1, R, a);
    st[p].resize(st[p << 1].size() + st[p << 1 |
        1].size());
    merge(all(st[p << 1]), all(st[p << 1 | 1]), st[
        p].begin());
}
int less_than(int p, int L, int R, int qL, int qR
    , ll val) {
    if (qL > R || qR < L) return 0;
    if (qL <= L && R <= qR) return lower_bound(all(
        st[p]), val) - st[p].begin();
    int mid = (L + R) / 2;
    return less_than(p << 1, L, mid, qL, qR, val) +
        less_than(p << 1 | 1, mid + 1, R, qL, qR, val)
        ;
}
int less_equal(int p, int L, int R, int qL, int
    qR, ll val) {
    if (qL > R || qR < L) return 0;
    if (qL <= L && R <= qR) return upper_bound(all(
        st[p]), val) - st[p].begin();
    int mid = (L + R) / 2;
    return less_equal(p << 1, L, mid, qL, qR, val)
        + less_equal(p << 1 | 1, mid + 1, R, qL, qR,
            val);
}
int count_less_than(int L, int R, ll val) {
    return less_than(1, 0, n - 1, L, R, val); }
int count_less_equal(int L, int R, ll val) {
    return less_equal(1, 0, n - 1, L, R, val); }
};

/**
 * Uso[0]: SqrtDecomposition sq(a); sq.update(pos,
 * val); sq.query(L, R);
 * Descomposicion en bloques para sumas de rangos
 * con actualizaciones puntuales.
 * Complejidad: actualizacion y consulta  $O(\sqrt{N})$ .
 */
struct SqrtDecomposition {
    int n, block_size;
    vll a, blocks;
    SqrtDecomposition(const vll& arr) : n(arr.size())
        , block_size(max(1, (int)sqrt(n))), a(arr) {
        blocks.assign((n + block_size - 1) / block_size
            , 0);
        for (int i = 0; i < n; i++) blocks[i /
            block_size] += a[i];
    }
    void update(int pos, ll val) {
        int b = pos / block_size;
        a[pos] = val;
        blocks[b] = 0;
        int end = min(n, (b + 1) * block_size);
        for (int i = b * block_size; i < end; i++)
            blocks[b] += a[i];
    }
    ll query(int L, int R) {
        int bL = L / block_size, bR = R / block_size;
        ll res = 0;
        if (bL == bR) {

```

```

    for (int i = L; i <= R; i++) res += a[i];
    return res;
}
for (int i = L; i < (bL + 1) * block_size; i++)
    res += a[i];
for (int b = bL + 1; b < bR; b++) res += blocks[b];
for (int i = bR * block_size; i <= R; i++) res += a[i];
return res;
}
};

mt19937_64 rng(chrono::steady_clock::now().
    time_since_epoch().count());
struct TreapNode {
    ll key, prior;
    int sz;
    TreapNode *l, *r;
    TreapNode(ll key) : key(key), prior(rng()), sz(1)
        , l(nullptr), r(nullptr) {}
};
typedef TreapNode* pNode;
/**
 * Uso[0]: Treap t; t.insert(x); t.erase(x); t.
 * kth_element(k); t.count_less_than(x);
 * Arbol binario de busqueda balanceado
 * aleatoriamente (treap) con consultas por orden.
 * Complejidad: cada operacion O(log N) esperado.
 */
struct Treap {
    pNode root = nullptr;
    int sz(pNode t) { return t ? t->sz : 0; }
    void upd(pNode t) { if (t) t->sz = 1 + sz(t->l) +
        sz(t->r); }
    void split(pNode t, ll key, pNode& l, pNode& r) {
        // l: keys <= key
        if (!t) { l = r = nullptr; return; }
        if (key >= t->key) split(t->r, key, t->r, r), l
            = t;
        else split(t->l, key, l, t->l), r = t;
        upd(t);
    }
    void merge(pNode& t, pNode l, pNode r) {
        if (!l || !r) { t = l ? l : r; return; }
        if (l->prior > r->prior) merge(l->r, l->r, r),
            t = l;
        else merge(r->l, l, r->l), t = r;
        upd(t);
    }
    void insert(ll key) {
        pNode l, r;
        split(root, key, l, r);
        pNode node = new TreapNode(key);
        merge(l, l, node);
        merge(root, l, r);
    }
    void erase(ll key) {
        pNode l, m, r;
        split(root, key - 1, l, r);
        split(r, key, m, r);
        if (m) {
            pNode tmp = m;
            merge(m, m->l, m->r);
            delete tmp;
        }
    }
};

```

```

merge(root, l, m);
merge(root, root, r);
}
ll kth_element(int k) { // k-esimo menor (0-
    indexed). LINF si fuera de rango
    pNode cur = root;
    while (cur) {
        int ls = sz(cur->l);
        if (ls == k) return cur->key;
        if (ls > k) cur = cur->l;
        else k -= ls + 1, cur = cur->r;
    }
    return LINF;
}
int count_less_than(ll key) {
    pNode l, r;
    split(root, key - 1, l, r);
    int res = sz(l);
    merge(root, l, r);
    return res;
}
};
/**
 * Uso[0]: ImplicitTreap it; it.insert(pos, val);
 * it.erase(pos); it.query_range(L, R); it.
 * reverse_range(L, R);
 * Treap implicito para secuencias dinamicas con
 * insercion/borrado por posicion, suma de rangos
 * y reversiones perezosas.
 * Complejidad: cada operacion O(log N) esperado.
 */
struct ImplicitNode {
    ll val, sum, prior;
    int sz;
    bool lazy_rev;
    ImplicitNode *l, *r;
    ImplicitNode(ll val) : val(val), sum(val), prior(
        rng()), sz(1), lazy_rev(false), l(nullptr), r(
        nullptr) {}
};
typedef ImplicitNode* pINode;
struct ImplicitTreap {
    pINode root = nullptr;
    int sz(pINode t) { return t ? t->sz : 0; }
    ll sum(pINode t) { return t ? t->sum : 0; }
    void push(pINode t) {
        if (t && t->lazy_rev) {
            t->lazy_rev = false;
            swap(t->l, t->r);
            if (t->l) t->l->lazy_rev ^= true;
            if (t->r) t->r->lazy_rev ^= true;
        }
    }
    void upd(pINode t) {
        if (t) t->sz = 1 + sz(t->l) + sz(t->r), t->sum
            = t->val + sum(t->l) + sum(t->r);
    }
    void split(pINode t, int k, pINode& l, pINode& r)
        { // l: primeros k elementos
        if (!t) { l = r = nullptr; return; }
        push(t);
        int implicit = sz(t->l) + 1;
        if (k >= implicit) split(t->r, k - implicit, t
            ->r, r), l = t;
        else split(t->l, k, l, t->l), r = t;
    }
};

```

```

    upd(t);
}
void merge(pINode& t, pINode l, pINode r) {
    push(1); push(r);
    if (!l || !r) { t = l ? l : r; return; }
    if (l->prior > r->prior) merge(l->r, l->r, r),
        t = l;
    else merge(r->l, l, r->l), t = r;
    upd(t);
}
void insert(int pos, ll val) {
    pINode l, r;
    split(root, pos, l, r);
    pINode node = new ImplicitNode(val);
    merge(l, l, node);
    merge(root, l, r);
}
void erase(int pos) {
    pINode l, mid, r;
    split(root, pos, l, r);
    split(r, 1, mid, r);
    delete mid;
    merge(root, l, r);
}
void reverse_range(int L, int R) {
    pINode l, mid, r;
    split(root, L, l, mid);
    split(mid, R - L + 1, mid, r);
    mid->lazy_rev ^= true;
    merge(root, l, mid);
    merge(root, root, r);
}
ll query_range(int L, int R) {
    pINode l, mid, r;
    split(root, L, l, mid);
    split(mid, R - L + 1, mid, r);
    ll ans = sum(mid);
    merge(root, l, mid);
    merge(root, root, r);
    return ans;
}
};

/**
 * Uso[0]: Mo mo(n); mo.add_query(1, r); auto ans =
 * mo.solve(add, remove, get_ans);
 * Algoritmo de Mo 1D para responder consultas de
 * rango offline cuando add y remove son O(1).
 * Complejidad: O((N+Q)√N) movimientos.
 */
struct Mo {
    struct Query { int l, r, id, block; };
    int n, bs;
    vector<Query> qs;
    Mo(int n, int bs = 0) : n(n), bs(bs ? bs : max(1,
        (int)sqrt(n))) {}
    void add_query(int l, int r) { qs.pb({l, r, (int)
        qs.size(), 1 / bs}); }
    template <class Add, class Remove, class GetAns>
    vector<ll> solve(Add add, Remove remove, GetAns
        get_ans) {
        vector<ll> res(qs.size());
        sort(all(qs), [](const Query& a, const Query& b) {
            if (a.block != b.block) return a.block < b.
                block;

```

```

                return (a.block & 1) ? a.r > b.r : a.r < b.r;
                // odd-even para acelerar
            });
            int cur_l = 0, cur_r = -1;
            for (const Query& q : qs) {
                while (cur_l > q.l) add(--cur_l);
                while (cur_r < q.r) add(++cur_r);
                while (cur_l < q.l) remove(cur_l++);
                while (cur_r > q.r) remove(cur_r--);
                res[q.id] = get_ans();
            }
            return res;
        }
    };

    /**
     * Uso[0]: MoWithUpdates mo(n); mo.add_update(pos,
     * antes, despues); mo.add_query(1, r); auto ans =
     * mo.solve(add, remove, apply, revert, get_ans);
     * Algoritmo de Mo con actualizaciones puntuales (
     * Mo 3D); responde consultas sobre el arreglo en
     * un instante dado.
     * Complejidad: O((N+Q)5/3) movimientos esperados.
     */
    struct MoWithUpdates {
        struct Query { int l, r, t, id; };
        struct Update { int pos; ll old_val, new_val; };
        int n, bs;
        vector<Query> qs;
        vector<Update> ups;
        MoWithUpdates(int n, int bs = 0) : n(n), bs(bs ?
            bs : max(1, (int)pow(n, 2.0 / 3.0))) {}
        void add_update(int pos, ll old_val, ll new_val)
            { ups.pb({pos, old_val, new_val}); }
        void add_query(int l, int r) { qs.pb({l, r, (int)
            ups.size(), (int)qs.size()}); }
        template <class Add, class Remove, class Apply,
            class Revert, class GetAns>
        vector<ll> solve(Add add, Remove remove, Apply
            apply, Revert revert, GetAns get_ans) {
            vector<ll> res(qs.size());
            sort(all(qs), [&](const Query& a, const Query&
                b) {
                int ba = a.l / bs, bb = b.l / bs;
                if (ba != bb) return ba < bb;
                int ca = a.r / bs, cb = b.r / bs;
                if (ca != cb) return ca < cb;
                return a.t < b.t;
            });
            int cur_l = 0, cur_r = -1, cur_t = 0;
            for (const Query& q : qs) {
                while (cur_t < q.t) { apply(ups[cur_t], cur_l
                    , cur_r); cur_t++; }
                while (cur_t > q.t) { cur_t--; revert(ups[
                    cur_t], cur_l, cur_r); }
                while (cur_l > q.l) add(--cur_l);
                while (cur_r < q.r) add(++cur_r);
                while (cur_l < q.l) remove(cur_l++);
                while (cur_r > q.r) remove(cur_r--);
                res[q.id] = get_ans();
            }
            return res;
        }
    };

    /**

```

```

* Uso[0]: RollbackMo mo(n); mo.add_query(1, r); mo
.solve(add, current, snapshot, rollback, reset)
;
* Variante de Mo con rollback para estructuras
donde agregar es  $O(1)$  pero eliminar es costoso
o imposible.
* Complejidad:  $O((N + Q)\sqrt{N})$  operaciones de add.
*/
struct RollbackMo {
    struct Query { int l, r, id; };
    int n, bs;
    vector<Query> qs;
    RollbackMo(int n, int bs = 0) : n(n), bs(bs ? bs
        : max(1, (int)sqrt(n))) {}
    void add_query(int l, int r) { qs.pb({l, r, (int)
        qs.size()}); }
    template <class Add, class Current, class
        Snapshot, class Rollback, class Reset>
    vector<ll> solve(Add add, Current current,
        Snapshot snapshot, Rollback rollback, Reset
        reset) {
        vector<ll> res(qs.size());
        sort(all(qs), [&](const Query& a, const Query&
            b) {
            if (a.l / bs != b.l / bs) return a.l / bs < b
                .l / bs;
            return a.r < b.r;
        });
        int block = -1, cur_r = -1;
        for (const Query& q : qs) {
            int b = q.l / bs;
            int block_end = min(n - 1, (b + 1) * bs - 1);
            if (b != block) {
                block = b;
                cur_r = block_end; // rango persistente
                vacio
                reset();
            }
            if (q.r <= block_end) {
                // query contenida en el bloque: brute
                force sobre estructura vacia
                snapshot();
                for (int i = q.l; i <= q.r; i++) add(i);
                res[q.id] = current();
                rollback();
            } else {
                while (cur_r < q.r) add(++cur_r);
                snapshot();
                for (int i = block_end; i >= q.l; i--) add(
                    i);
                res[q.id] = current();
                rollback();
            }
        }
        return res;
    }
};

```

Strings

```

/**
* Uso: to_lower(s), to_upper(s), trim(s),
    replace_all(s, "a", "b"), split(s, ','), join(v
        , " ");
* Utilidades de procesamiento de strings:
    mayusculas/minusculas, recorte, reemplazo,
    division y union.

```

```

* Complejidad:  $O(N)$  por operacion.
*/
void to_lower(string& s) { transform(all(s), s.
    begin(), ::tolower); }
void to_upper(string& s) { transform(all(s), s.
    begin(), ::toupper); }
void trim(string& s) {
    s.erase(s.begin(), find_if(all(s), [](unsigned
        char c) { return !isspace(c); }));
    s.erase(find_if(rall(s), [](unsigned char c) {
        return !isspace(c); }).base(), s.end());
}
void replace_all(string& s, const string& old,
    const string& nw) {
    size_t pos = 0;
    while ((pos = s.find(old, pos)) != string::npos)
        s.replace(pos, old.size(), nw), pos += nw.size
            ();
}
vector<string> split(const string& s, char delim) {
    vector<string> res;
    string item;
    istringstream iss(s);
    while (getline(iss, item, delim)) res.pb(item);
    return res;
}
string join(const vector<string>& v, const string&
    delim) {
    string res;
    for (int i = 0; i < v.size(); i++) {
        res += v[i];
        if (i + 1 < v.size()) res += delim;
    }
    return res;
}
/**
* Uso: auto pi = prefix_function(s);
* Funcion prefijo de KMP:  $pi[i]$  es el prefijo propio
    mas largo que tambien es suñjo de  $s[0..i]$ .
* Complejidad:  $O(N)$ .
*/
vi prefix_function(const string& s) {
    int n = s.size();
    vi pi(n, 0);
    for (int i = 1; i < n; i++) {
        int j = pi[i - 1];
        while (j > 0 && s[i] != s[j]) j = pi[j - 1];
        if (s[i] == s[j]) j++;
        pi[i] = j;
    }
    return pi;
}
/**
* Uso: auto z = z_function(s);
* Funcion Z:  $z[i]$  es la longitud del prefijo de s que
    coincide con el suñjo que inicia en i.
* Complejidad:  $O(N)$ .
*/
vi z_function(const string& s) {
    int n = s.size(), l = 0, r = 0;
    vi z(n, 0);
    for (int i = 1; i < n; i++) {
        if (i <= r) z[i] = min(r - i + 1, z[i - 1]);
        while (i + z[i] < n && s[z[i]] == s[i + z[i]])

```

```

    z[i]++;
    if (i + z[i] - 1 > r) l = i, r = i + z[i] - 1;
}
return z;
}

```

```

/**
 * Uso: string res = remove_all_occurrences(s, p);
 * Elimina todas las apariciones de un patron p en
 * s, incluidas las encadenadas.
 * Complejidad:  $O(N + M)$  con  $N = |s|$  y  $M = |p|$ .
 */

```

```

string remove_all_occurrences(const string& s,
    const string& p) {
    vi pi = prefix_function(p);
    string res;
    vi state(s.size() + 1, 0);
    for (char c : s) {
        res += c;
        int j = state[res.size() - 1];
        while (j > 0 && p[j] != c) j = pi[j - 1];
        if (p[j] == c) j++;
        state[res.size()] = j;
        if (j == p.size()) res.resize(res.size() - p.size());
    }
    return res;
}

```

```

/**
 * Uso[0]: auto occ = rabin_karp(text, pattern);
 * Encuentra las posiciones donde comienza un
 * patron usando hashing (rolling hash).
 * Complejidad:  $O(T + S)$  con  $T = |text|$  y  $S = |pattern|$ .
 */

```

```

vi rabin_karp(const string& text, const string&
    pattern) {
    const ll P = 31, M = 1e9 + 9;
    int S = pattern.size(), T = text.size();
    if (!S || S > T) return {};
    vll p_pow(max(S, T));
    p_pow[0] = 1;
    for (int i = 1; i < p_pow.size(); i++) p_pow[i] =
        p_pow[i - 1] * P % M;
    vll h(T + 1, 0);
    for (int i = 0; i < T; i++) h[i + 1] = (h[i] + (
        text[i] - 'a' + 1) * p_pow[i]) % M;
    ll hs = 0;
    for (int i = 0; i < S; i++) hs = (hs + (pattern[i]
        - 'a' + 1) * p_pow[i]) % M;
    vi occ;
    for (int i = 0; i + S - 1 < T; i++) {
        ll cur = (h[i + S] + M - h[i]) % M;
        if (cur == hs * p_pow[i] % M) occ.pb(i);
    }
    return occ;
}

```

```

/**
 * Uso: auto p = manacher(s);
 * Radios de los palindromos en cada posicion (
 * incluyendo centros pares); el maximo da el
 * palindromo mas largo.
 * Complejidad:  $O(N)$ .
 */
vi manacher(const string& s) {
    string t = "^#";

```

```

    for (char c : s) t += c, t += '#';
    t += '$';
    int n = t.size(), c = 0, r = 0;
    vi p(n, 0);
    for (int i = 1; i < n - 1; i++) {
        int mir = 2 * c - i;
        p[i] = r > i ? min(r - i, p[mir]) : 0;
        while (t[i + 1 + p[i]] == t[i - 1 - p[i]]) p[i]
            ++;
        if (i + p[i] > r) c = i, r = i + p[i];
    }
    return vi(p.begin() + 1, p.end() - 1);
}

```

```

/**
 * Uso[0]: StringHash sh(s); sh.get_hash(l, r);
 * Hash doble de substrings para compararlos en
 *  $O(1)$ .
 * Complejidad: preproceso  $O(N)$ , consulta  $O(1)$ .
 */

```

```

struct StringHash {
    static constexpr ll MOD1 = 1000000007LL; // 1e9 +
    7
    static constexpr ll MOD2 = 1000000009LL; // 1e9 +
    9
    static constexpr ll BASE1 = 31;
    static constexpr ll BASE2 = 37;
    string s;
    int n;
    vll h1, h2, p1, p2;
    StringHash(const string& s) : s(s), n(s.size()),
        h1(n + 1, 0), h2(n + 1, 0), p1(n + 1, 1), p2(n
        + 1, 1) {
        for (int i = 0; i < n; i++) {
            h1[i + 1] = (h1[i] * BASE1 + (s[i] - 'a' + 1)
            ) % MOD1;
            h2[i + 1] = (h2[i] * BASE2 + (s[i] - 'a' + 1)
            ) % MOD2;
            p1[i + 1] = p1[i] * BASE1 % MOD1;
            p2[i + 1] = p2[i] * BASE2 % MOD2;
        }
    }
    uint64_t get_hash(int l, int r) {
        ll x1 = (h1[r + 1] - h1[l] * p1[r - l + 1] %
        MOD1 + MOD1) % MOD1;
        ll x2 = (h2[r + 1] - h2[l] * p2[r - l + 1] %
        MOD2 + MOD2) % MOD2;
        return (uint64_t)x1 << 32 | (uint32_t)x2;
    }
};

```

```

/**
 * Uso: Trie t; t.insert(s); t.search(s);
 * Trie (arbol de prefijos) sobre el alfabeto de
 * minúsculas con insercion y busqueda de palabras
 * .
 * Complejidad: insercion y busqueda  $O(L)$  con  $L$  la
 * longitud de la cadena.
 */

```

```

struct Trie {
    struct Node { int next[26] = {}; bool is_end =
        false; int count = 0; };
    vector<Node> t;
    Trie() { t.emplace_back(); }
    void insert(const string& s) {
        int v = 0;
        for (char ch : s) {

```

```

    int c = ch - 'a';
    if (!t[v].next[c]) t[v].next[c] = t.size(), t
.emplace_back();
    v = t[v].next[c];
    t[v].count++;
}
t[v].is_end = true;
}
bool search(const string& s) {
    int v = 0;
    for (char ch : s) {
        int c = ch - 'a';
        if (!t[v].next[c]) return false;
        v = t[v].next[c];
    }
    return t[v].is_end;
}
};

/**
 * Uso[0]: AhoCorasick ac; ac.insert(p); ac.build()
; ac.count_occurrences(text); ac.
first_occurrences(text);
 * Automata Aho-Corasick para buscar multiples
patrones en un texto: conteo y primera
aparicion de cada uno.
 * Complejidad: construccion  $O(\sum |p_i| \cdot 26)$ ,
procesamiento del texto  $O(T)$ .
 */
struct AhoCorasick {
    struct Node {
        int next[26] = {};
        int link = 0;
        vi word_ids;
    };
    vector<Node> t;
    vi bfs_order, pat_len;
    int num_patterns = 0;
    AhoCorasick() { t.emplace_back(); }
    int insert(const string& s) {
        int v = 0, id = num_patterns++;
        pat_len.pb(s.size());
        for (char ch : s) {
            int c = ch - 'a';
            if (!t[v].next[c]) t[v].next[c] = t.size(), t
.emplace_back();
            v = t[v].next[c];
        }
        t[v].word_ids.pb(id);
        return id;
    }
    void build() {
        queue<int> q;
        for (int c = 0; c < 26; c++) {
            if (t[0].next[c]) q.push(t[0].next[c]);
            else t[0].next[c] = 0;
        }
        while (!q.empty()) {
            int u = q.front(); q.pop();
            bfs_order.pb(u);
            for (int c = 0; c < 26; c++) {
                int v = t[u].next[c];
                if (v) t[v].link = t[t[u].link].next[c], q.
push(v);
                else t[u].next[c] = t[t[u].link].next[c];
            }
        }
    }
};

```

```

    }
}
// Numero de ocurrencias de cada patron en text
vi count_occurrences(const string& text) {
    vi freq(t.size(), 0);
    int u = 0;
    for (char ch : text) u = t[u].next[ch - 'a'],
freq[u]++;
    for (int i = bfs_order.size() - 1; i >= 0; i--)
    {
        int cur = bfs_order[i];
        freq[t[cur].link] += freq[cur];
    }
    vi ans(num_patterns, 0);
    for (int i = 1; i < t.size(); i++) if (freq[i])
        for (int id : t[i].word_ids) ans[id] += freq[i]
    };
    return ans;
}
// Posicion de la primera ocurrencia de cada
patron (o -1)
vi first_occurrences(const string& text) {
    vi first(t.size(), INF);
    int u = 0;
    for (int i = 0; i < text.size(); i++) {
        u = t[u].next[text[i] - 'a'];
        first[u] = min(first[u], i);
    }
    for (int i = bfs_order.size() - 1; i >= 0; i--)
    {
        int cur = bfs_order[i];
        first[t[cur].link] = min(first[t[cur].link],
first[cur]);
    }
    vi ans(num_patterns, -1);
    for (int i = 1; i < t.size(); i++) if (first[i]
!= INF)
        for (int id : t[i].word_ids) {
            int st = first[i] - pat_len[id] + 1;
            ans[id] = ans[id] == -1 ? st : min(ans[id],
st);
        }
    return ans;
}
};

/**
 * Uso[0]: SuffixAutomaton sam(s); sam.
distinct_substrings(); sam.count_occurrences(p)
; sam.first_occurrence(p);
 * Automata de sufijos: substrings distintos,
ocurrencias y primera aparicion de un patron.
 * Complejidad: construccion  $O(N \cdot 26)$ , consultas
 $O(|p|)$ .
 */
struct SuffixAutomaton {
    struct Node {
        int len = 0, link = -1, next[26] = {};
        int firstpos = -1;
        ll cnt = 0;
    };
    vector<Node> t;
    int last;
    bool occ_computed = false;
    SuffixAutomaton(const string& s = "") {
        t.reserve(s.size() * 2 + 1);
    }
};

```

```

    t.emplace_back();
    last = 0;
    for (char c : s) extend(c - 'a');
}
void extend(int c) {
    int cur = t.size();
    t.emplace_back();
    t[cur].len = t[last].len + 1;
    t[cur].firstpos = t[cur].len - 1;
    t[cur].cnt = 1;
    int p = last;
    while (p != -1 && !t[p].next[c]) t[p].next[c] =
        cur, p = t[p].link;
    if (p == -1) t[cur].link = 0;
    else {
        int q = t[p].next[c];
        if (t[p].len + 1 == t[q].len) t[cur].link = q;
        else {
            int clone = t.size();
            t.push_back(t[q]);
            t[clone].len = t[p].len + 1;
            t[clone].cnt = 0;
            while (p != -1 && t[p].next[c] == q) t[p].
                next[c] = clone, p = t[p].link;
            t[q].link = t[cur].link = clone;
        }
    }
    last = cur;
}
void compute_occurrences() {
    if (occ_computed) return;
    occ_computed = true;
    int n = t.size();
    vi c(n + 1, 0), order(n);
    for (int i = 0; i < n; i++) c[t[i].len]++;
    for (int i = 1; i <= n; i++) c[i] += c[i - 1];
    for (int i = 0; i < n; i++) order[--c[t[i].len]] = i;
    for (int i = n - 1; i > 0; i--) {
        int u = order[i];
        if (t[u].link != -1) t[t[u].link].cnt += t[u].cnt;
    }
}
ll distinct_substrings() {
    ll ans = 0;
    for (int i = 1; i < t.size(); i++) ans += t[i].len - t[t[i].link].len;
    return ans;
}
int first_occurrence(const string& p) {
    int u = 0;
    for (char ch : p) {
        int c = ch - 'a';
        if (!t[u].next[c]) return -1;
        u = t[u].next[c];
    }
    return t[u].firstpos - p.size() + 1;
}
ll count_occurrences(const string& p) {
    compute_occurrences();
    int u = 0;
    for (char ch : p) {
        int c = ch - 'a';
        if (!t[u].next[c]) return 0;

```

```

        u = t[u].next[c];
    }
    return t[u].cnt;
};

/**
 * Uso[0]: SuffixArray sa(s); sa.p (sufijos ordenados);
 * sa.get_lcp(i, j);
 * Arreglo de sufijos con tabla RMQ de lcp para consultar el prefijo comun mas largo de dos sufijos en O(1).
 * Complejidad: construccion O(N log N), consulta lcp O(1).
 */
struct SuffixArray {
    string s;
    int n, log_n;
    vi p, c, lcp, rank;
    vector<vi> st;
    SuffixArray(string s) : s(s + "$"), n(s.size()) {
        build_sa(); build_lcp(); build_rmq();
    }
    void build_sa() {
        const int alphabet = 256;
        p.assign(n, 0); c.assign(n, 0);
        vi cnt(max(alphabet, n), 0), p_new(n), c_new(n);
        for (int i = 0; i < n; i++) cnt[s[i]]++;
        for (int i = 1; i < alphabet; i++) cnt[i] += cnt[i - 1];
        for (int i = 0; i < n; i++) p[--cnt[s[i]]] = i;
        c[p[0]] = 0;
        int classes = 1;
        for (int i = 1; i < n; i++) {
            if (s[p[i]] != s[p[i - 1]]) classes++;
            c[p[i]] = classes - 1;
        }
        for (int k = 0; (1 << k) < n; k++) {
            for (int i = 0; i < n; i++) {
                p_new[i] = p[i] - (1 << k);
                if (p_new[i] < 0) p_new[i] += n;
            }
            fill(cnt.begin(), cnt.begin() + classes, 0);
            for (int i = 0; i < n; i++) cnt[c[p_new[i]]]++;
            for (int i = 1; i < classes; i++) cnt[i] += cnt[i - 1];
            for (int i = n - 1; i >= 0; i--) p[--cnt[c[p_new[i]]]] = p_new[i];
            c_new[p[0]] = 0;
            classes = 1;
            for (int i = 1; i < n; i++) {
                pair<int, int> cur = {c[p[i]], c[(p[i] + (1 << k)) % n]};
                pair<int, int> prev = {c[p[i - 1]], c[(p[i - 1] + (1 << k)) % n]};
                if (cur != prev) classes++;
                c_new[p[i]] = classes - 1;
            }
            c.swap(c_new);
        }
    }
    void build_lcp() {
        lcp.assign(n, 0);
        rank.assign(n, 0);

```

```

    for (int i = 0; i < n; i++) rank[p[i]] = i;
    int k = 0;
    for (int i = 0; i < n - 1; i++) {
        int pi = rank[i], j = p[pi - 1];
        while (s[i + k] == s[j + k]) k++;
        lcp[pi] = k;
        if (k) k--;
    }
}

void build_rmq() {
    log_n = log2(n) + 1;
    st.assign(n, vi(log_n));
    for (int i = 0; i < n; i++) st[i][0] = lcp[i];
    for (int j = 1; j < log_n; j++)
        for (int i = 0; i + (1 <= j) <= n; i++)
            st[i][j] = min(st[i][j - 1], st[i + (1 <= j) - 1][j - 1]);
}

// LCP de los sufijos que empiezan en i y j
int get_lcp(int i, int j) {
    if (i == j) return n - 1 - i;
    int u = rank[i], v = rank[j];
    if (u > v) swap(u, v);
    u++;
    int k = log2(v - u + 1);
    return min(st[u][k], st[v - (1 <= k) + 1][k]);
}
};

```

/**

* *Uso[0]: PalindromicTree pt(s); pt.
count_palindromes(); pt.compute_occurrences();*

* *Arbol de palindromos (eertree): todos los
palindromos distintos, sus ocurrencias y su
primera posicion.*

* *Complejidad: construccion $O(N \cdot 26)$, ocurrencias
 $O(N)$.*

*/

```

struct PalindromicTree {
    struct Node {
        int next[26] = {};
        int len = 0, link = 0;
        ll occ = 0;
        int first_pos = -1; // posicion donde aparece
        // por primera vez
    };
    vector<Node> t;
    string s; // s[0] = sentinel, nunca coincide con
    // una letra
    int last;
    PalindromicTree(const string& str = "") {
        t.emplace_back(); t.emplace_back();
        t[0].len = -1; t[0].link = 0; // raiz
        // imaginaria (len -1)
        t[1].len = 0; t[1].link = 0; // raiz del vacio
        // (len 0)
        last = 1;
        s = "$";
        for (char c : str) add(c);
    }
    int get_link(int v, int pos) {
        while (true) {
            int idx = pos - t[v].len - 1;
            if (idx >= 0 && s[idx] == s[pos]) break;
            v = t[v].link;
        }
    }
}

```

```

    return v;
}

void add(char c) {
    s += c;
    int pos = s.size() - 1;
    int cur = get_link(last, pos);
    int idx = c - 'a';
    if (!t[cur].next[idx]) {
        int now = t.size();
        t.emplace_back();
        t[now].len = t[cur].len + 2;
        t[now].first_pos = pos - t[now].len; // 0-
        // indexado
        t[now].link = t[now].len == 1 ? 1 : t[
            get_link(t[cur].link, pos)].next[idx];
        t[cur].next[idx] = now;
    }
    last = t[cur].next[idx];
    t[last].occ++;
}

ll count_palindromes() { return t.size() - 2; }
// palindromos distintos
void compute_occurrences() { // occ[u] = veces
    // que aparece el palindromo u
    for (int i = t.size() - 1; i > 1; i--) t[t[i].
        link].occ += t[i].occ;
}
};

```

Dynamic Programming

/**

* *Uso: LiChaoTree cht; cht.add(m, c); ll v = cht.
query(x);*

* *Arbol de Li Chao para consultar el minimo de
 $m \cdot x + c$ entre las rectas insertadas (cambiar <
por > para max).*

* *Complejidad: insercion y consulta $O(\log R)$ con R
el tamano del rango de x.*

*/

```

struct LiChaoTree {
    struct Line {
        ll m, c;
        Line(ll m = 0, ll c = LINF) : m(m), c(c) {}
        ll eval(ll x) const { return m * x + c; }
    };
    struct Node { Line line; int lc = -1, rc = -1; };
    vector<Node> st;
    ll min_x, max_x;
    LiChaoTree(ll mn = -1e9, ll mx = 1e9) : min_x(mn),
        max_x(mx) { st.emplace_back(); }
    void add_line(Line nw, int v, ll l, ll r) {
        ll mid = l + (r - l) / 2;
        bool left = nw.eval(l) < st[v].line.eval(l);
        bool midb = nw.eval(mid) < st[v].line.eval(mid);
        if (midb) swap(st[v].line, nw);
        if (l == r) return;
        if (left != midb) {
            if (st[v].lc == -1) st[v].lc = st.size(), st.
                emplace_back();
            add_line(nw, st[v].lc, l, mid);
        } else {
            if (st[v].rc == -1) st[v].rc = st.size(), st.
                emplace_back();
            add_line(nw, st[v].rc, mid + 1, r);
        }
    }
}

```

```

    }
}
void add(ll m, ll c) { add_line(Line(m, c), 0,
    min_x, max_x); }
ll query(ll x, int v, ll l, ll r) {
    if (v == -1) return LINF;
    ll res = st[v].line.eval(x);
    if (l == r) return res;
    ll mid = l + (r - l) / 2;
    if (x <= mid) return min(res, query(x, st[v].lc,
        l, mid));
    return min(res, query(x, st[v].rc, mid + 1, r));
}
ll query(ll x) { return query(x, 0, min_x, max_x); }
};

/**
 * Uso: dnc_dp(1, n, 0, n, dp_prev, dp_curr, cost);
 * Divide & conquer DP: resuelve
 *  $dp[i][j] = \min_{k < j} (dp[i-1][k] + cost(k+1, j))$  cuando la
 * decision optima es monotona.
 * Complejidad:  $O(KN \log N)$  con cost evaluada en  $O(1)$ .
 */
template<class CostFunc>
void dnc_dp(int l, int r, int optl, int optr, const
    vll& prev, vll& curr, CostFunc& cost) {
    if (l > r) return;
    int mid = (l + r) / 2;
    pair<ll, int> best = {LINF, -1};
    for (int k = optl; k <= min(mid - 1, optr); k++) {
        ll cur = prev[k] == LINF ? LINF : prev[k] +
            cost(k + 1, mid);
        if (cur < best.fi) best = {cur, k};
    }
    curr[mid] = best.fi;
    int opt = best.se == -1 ? optl : best.se;
    dnc_dp(l, mid - 1, optl, opt, prev, curr, cost);
    dnc_dp(mid + 1, r, opt, optr, prev, curr, cost);
}

/**
 * Uso: auto dp = knuth_dp(n, cost);
 * Optimizacion de Knuth para DP de intervalos
 *  $dp[i][j] = \min_k (dp[i][k] + dp[k][j]) + cost(i, j)$  con decision
 * monotona.
 * Complejidad:  $O(N^2)$ .
 */
template<class CostFunc>
vector<vll> knuth_dp(int n, CostFunc& cost) {
    vector<vll> dp(n + 1, vll(n + 1, 0));
    vector<vi> opt(n + 1, vi(n + 1, 0));
    for (int i = 0; i < n; i++) opt[i][i + 1] = i;
    for (int len = 2; len <= n; len++) {
        for (int i = 0; i + len <= n; i++) {
            int j = i + len;
            dp[i][j] = LINF;
            for (int k = opt[i][j - 1]; k <= min(j - 1,
                opt[i + 1][j]); k++) {
                ll cur = dp[i][k] + dp[k][j] + cost(i, j);
                if (cur < dp[i][j]) dp[i][j] = cur, opt[i][
                    j] = k;
            }
        }
    }
}

```

```

    return dp;
}

/**
 * Uso: DigitDP::solve(R) - DigitDP::solve(L-1);
 * DP sobre digitos para contar numeros en  $[0, N]$  que
 * cumplen condiciones; ajustar memo y
 * transiciones segun el problema.
 * Complejidad:  $O(\text{digitos} \times \text{estados})$  por llamada.
 */
namespace DigitDP {
    string s;
    ll memo[20][2][2]; // agregar estados segun
    // problema
    ll dp(int idx, bool tight, bool lz) {
        if (idx == s.size()) return 1; // 1 si el
        // numero es valido
        if (memo[idx][tight][lz] != -1) return memo[idx]
            [tight][lz];
        int lim = tight ? s[idx] - '0' : 9;
        ll ans = 0;
        for (int d = 0; d <= lim; d++) {
            ans += dp(idx + 1, tight && d == lim, lz && d
                == 0);
            return memo[idx][tight][lz] = ans;
        }
    }
    ll solve(ll n) {
        if (n < 0) return 0;
        s = to_string(n);
        memset(memo, -1, sizeof(memo));
        return dp(0, true, true);
    }
}

/**
 * Uso: sos_dp(dp, n_bits);
 * Suma sobre subconjuntos  $dp[\text{mask}]$  acumula la suma
 * de todos sus subconjuntos (sum over subsets).
 * Complejidad:  $O(N \cdot 2^N)$ .
 */
void sos_dp(vll& dp, int n) {
    for (int i = 0; i < n; i++) {
        for (int mask = 0; mask < (1 << n); mask++) {
            if (mask & (1 << i)) dp[mask] += dp[mask ^ (1
                << i)];
        }
    }
}

/**
 * Uso: auto [len, seq] = lis(a);
 * Subsecuencia creciente mas larga estricta, con
 * reconstruccion de la secuencia.
 * Complejidad:  $O(N \log N)$ .
 */
pair<int, vi> lis(const vi& a) {
    int n = a.size();
    vi d, pos, parent(n, -1);
    for (int i = 0; i < n; i++) {
        auto it = lower_bound(all(d), a[i]);
        int idx = it - d.begin();
        if (it == d.end()) d.pb(a[i]), pos.pb(i);
        else *it = a[i], pos[idx] = i;
        if (idx > 0) parent[i] = pos[idx - 1];
    }
    vi seq;
    for (int cur = pos.empty() ? -1 : pos.back(); cur
        != -1; cur = parent[cur]) seq.pb(a[cur]);
}

```

```

    reverse(all(seq));
    return {d.size(), seq};
}

/**
 * Uso: bitset_knapsack(pesos, target);
 * Determina si un subconjunto de pesos suma
 * exactamente un valor objetivo mediante bitsets.
 * Complejidad:  $O(N \cdot S/64)$  con  $S$  la suma maxima ( $10^5$ 
 * fija en MAX_SUM).
 */
const int MAX_SUM = 1e5 + 5;
bool bitset_knapsack(const vi& weights, int target)
{
    bitset<MAX_SUM> dp;
    dp[0] = 1;
    for (int w : weights) dp |= dp << w;
    return dp[target];
}

/**
 * Uso: ll ans = WQS::solve(check, K);
 * Búsqueda binaria de Lagrange (Alien's trick)
 * para optimizar un DP con la restricción de usar
 * exactamente K elementos o particiones.
 * Complejidad:  $O(\log R \cdot f)$  con  $f$  el costo del DP por
 * evaluación y  $R$  el rango de la penalidad.
 */
namespace WQS {
    template <class F>
    ll solve(F check, int K) {
        ll lo = -LINF, hi = LINF;
        while (lo + 1 < hi) {
            ll mid = lo + (hi - lo) / 2;
            if (check(mid).se >= K) lo = mid;
            else hi = mid;
        }
        auto [c, k] = check(lo);
        return c - lo * (ll)K; // quita la penalidad de
        los K elementos
    }
}

/**
 * Uso: CHTDeque cht; cht.add(m, c); cht.query(x);
 * // y cht.query_bs(x) para x arbitraria
 * Convex hull trick con deque para minimos  $m \cdot x + c$ 
 * con pendientes monotonas y consultas con x
 * creciente.
 * Complejidad: insercion y consulta amortizadas
 *  $O(1)$ ; consulta binaria  $O(\log N)$ .
 */
struct CHTDeque {
    struct Line { ll m, c; };
    deque<Line> dq;
    int slope_inc = 0; // +1 pendientes crecientes,
    -1 decrecientes (auto-detectado)
    static ll eval(const Line& l, ll x) { return l.m
        * x + l.c; }
    // true si l2 queda obsoleta entre l1 y l3
    bool is_bad(const Line& l1, const Line& l2, const
        Line& l3) {
        if (slope_inc > 0) return (__int128)(l1.c - l2.
            c) * (l3.m - l2.m) <= (__int128)(l2.c - l3.c) *
            (l2.m - l1.m);
        return (__int128)(l3.c - l1.c) * (l1.m - l2.m)
            <= (__int128)(l2.c - l1.c) * (l1.m - l3.m);
    }
};

```

```

}
void add(ll m, ll c) {
    if (!dq.empty() && dq.back().m == m) {
        if (dq.back().c <= c) return; // la nueva no
        aporta (min)
        dq.pop_back();
    }
    if (dq.size() >= 2 && slope_inc == 0) slope_inc
        = dq.back().m < m ? 1 : -1;
    Line nw{m, c};
    while (dq.size() >= 2 && is_bad(dq[dq.size() -
        2], dq.back(), nw)) dq.pop_back();
    dq.push_back(nw);
}
ll query(ll x) { // x debe ser no decreciente
    if (slope_inc > 0) { // pendientes crecientes:
        optimo va del fondo al frente
        while (dq.size() >= 2 && eval(dq.back(), x)
            >= eval(dq[dq.size() - 2], x)) dq.pop_back();
        return eval(dq.back(), x);
    }
    while (dq.size() >= 2 && eval(dq.front(), x) >=
        eval(dq[1], x)) dq.pop_front();
    return eval(dq.front(), x);
}
ll query_bs(ll x) { // x arbitraria
    int lo = 0, hi = (int)dq.size() - 1;
    while (lo < hi) {
        int mid = (lo + hi) / 2;
        if (eval(dq[mid], x) <= eval(dq[mid + 1], x))
            hi = mid;
        else lo = mid + 1;
    }
    return eval(dq[lo], x);
}
};

```

Mathematics

```

namespace MathAlgo {
    /**
     * Uso: modpow(a, b, mod), modinv(a, mod),
     * ext_gcd(a, b, x, y);
     * Potencia modular, inverso modular y euclides
     * extendido ( $ax + by = \gcd(a, b)$ ).
     * Complejidad:  $O(\log b)$  por operacion.
     */
    ll modpow(ll a, ll b, ll m = MOD) {
        ll r = 1; a %= m;
        while (b) { if (b & 1) r = r * a % m; a = a * a
            % m; b >>= 1; }
        return r;
    }
    // Euclides extendido: resuelve  $ax + by = \gcd(a, b)$ 
    ll ext_gcd(ll a, ll b, ll& x, ll& y) {
        if (!b) { x = 1; y = 0; return a; }
        ll x1, y1, d = ext_gcd(b, a % b, x1, y1);
        x = y1; y = x1 - y1 * (a / b); return d;
    }
    /**
     * Uso: auto [x, lcm] = modinv(a, mod);
     * Inverso modular de a modulo m (exige
     *  $\gcd(a, m) = 1$ ).
     * Complejidad:  $O(\log m)$ .
     */
}

```

```

11 modinv(11 a, 11 m = MOD) {
    11 x, y; assert(ext_gcd(a, m, x, y) == 1);
    return (x % m + m) % m;
}
/**
 * Uso: auto [x, lcm] = crt(a1, m1, a2, m2);
 * Teorema chino del resto para dos congruencias;
 * devuelve {-1, -1} si no tienen solución.
 * Complejidad:  $O(\log \min(m_1, m_2))$ .
 */
pair<11, 11> crt(11 a1, 11 m1, 11 a2, 11 m2) {
    11 p, q, g = ext_gcd(m1, m2, p, q);
    if (a1 % g != a2 % g) return {-1, -1};
    11 lcm = (m1 / g) * m2;
    11 res = a1 + (11)((__int128_t)(a2 - a1) / g *
    p % (m2 / g)) * m1;
    return {(res % lcm + lcm) % lcm, lcm};
}

/**
 * Uso: auto [x, lcm] = solve_crt_array(A, M);
 * Teorema chino del resto para un arreglo de
 * congruencias; devuelve {-1, -1} si alguna es
 * inconsistente.
 * Complejidad:  $O(K \log m)$  con  $K$  el número de
 * congruencias.
 */
pair<11, 11> solve_crt_array(const vector<11>& A,
    const vector<11>& M) {
    11 ans = A[0];
    11 lcm = M[0];
    for (int i = 1; i < A.size(); i++) {
        auto res = crt(ans, lcm, A[i], M[i]);
        if (res.first == -1) return {-1, -1}; //
        // No hay solución
        ans = res.first;
        lcm = res.second;
    }
    return {ans, lcm};
}

/**
 * Uso: Sieve sv(n); sv.primes; sv.spf; sv.mu; sv
 * .phi; sv.factorize(x);
 * Criba lineal con lista de primos, menor factor
 * primo, función de Mobius y totient de Euler.
 * Complejidad: criba  $O(N)$ , factorización de un
 * valor  $O(\log x)$ .
 */
struct Sieve {
    vector<int> primes, spf, mu, phi;
    Sieve(int n) {
        spf.assign(n + 1, 0); mu.assign(n + 1, 0);
        phi.assign(n + 1, 0);
        mu[1] = phi[1] = 1;
        for (int i = 2; i <= n; i++) {
            if (!spf[i]) spf[i] = i, primes.pb(i), mu[i]
            = -1, phi[i] = i - 1;
            for (int p : primes) {
                if (p > spf[i] || 1LL * i * p > n) break;
                spf[i * p] = p;
                if (i % p == 0) { mu[i * p] = 0; phi[i *
                p] = phi[i] * p; break; }
                mu[i * p] = -mu[i]; phi[i * p] = phi[i] *
                (p - 1);
            }
        }
    }
}

}

vector<int> factorize(int x) { // factores
    primos con multiplicidad
    vector<int> r;
    while (x > 1) r.pb(spf[x]), x /= spf[x];
    return r;
}

/**
 * Uso: Combinatorics c(n, mod); c.nCr(n, r); c.
    catalan(n);
 * Factoriales e inversos para combinar y números
    de Catalan modulo primo.
 * Complejidad: preproceso  $O(N)$ , consulta  $O(1)$ .
 */
struct Combinatorics {
    vector<11> fact, invfact;
    Combinatorics(int n, 11 mod = MOD) {
        fact.assign(n + 1, 1); invfact.assign(n + 1,
        1);
        for (int i = 1; i <= n; i++) fact[i] = fact[i
        - 1] * i % mod;
        invfact[n] = modinv(fact[n], mod);
        for (int i = n - 1; i >= 0; i--) invfact[i] =
        invfact[i + 1] * (i + 1) % mod;
        this->mod = mod;
    }
    11 mod;
    11 nCr(int n, int r) {
        if (r < 0 || r > n) return 0;
        return fact[n] * invfact[r] % mod * invfact[n
        - r] % mod;
    }
    11 catalan(int n) { return nCr(2 * n, n) *
    modinv(n + 1, mod) % mod; }
};

/**
 * Uso: gauss(matriz_extendida, soluciones);
 * Eliminación gaussiana con pivoteo para
    sistemas lineales; devuelve 1, INF o 0 según el
    número de soluciones.
 * Complejidad:  $O(n \cdot m^2)$ .
 */
int gauss(vector<vector<double>> a, vector<double>
    & ans) {
    int n = a.size(), m = a[0].size() - 1;
    vector<int> where(m, -1);
    for (int col = 0, row = 0; col < m && row < n;
    col++) {
        int sel = row;
        for (int i = row; i < n; i++) if (fabs(a[i][
        col]) > fabs(a[sel][col])) sel = i;
        if (fabs(a[sel][col]) < EPS) continue;
        for (int i = col; i <= m; i++) swap(a[sel][i]
        ], a[row][i]);
        where[col] = row;
        for (int i = 0; i < n; i++) if (i != row) {
            double c = a[i][col] / a[row][col];
            for (int j = col; j <= m; j++) a[i][j] -= a
            [row][j] * c;
        }
        row++;
    }
    ans.assign(m, 0);
}

```

```

    for (int i = 0; i < m; i++) if (where[i] != -1)
        ans[i] = a[where[i]][m] / a[where[i]][i];
    for (int i = 0; i < n; i++) {
        double s = 0;
        for (int j = 0; j < m; j++) s += ans[j] * a[i][j];
        if (fabs(s - a[i][m]) > EPS) return 0;
    }
    for (int i = 0; i < m; i++) if (where[i] == -1)
        return INF;
    return 1;
}

/**
 * Uso: fft(vec_complejos, invert); auto c =
 * multiply(a, b);
 * Transformada rapida de Fourier iterativa y
 * multiplicacion exacta de polinomios con
 * coeficientes enteros.
 * Complejidad:  $O(N \log N)$  con  $N$  potencia de dos.
 */
using cd = complex<double>;
// FFT iterativa, invert = true para transformada
// inversa
void fft(vector<cd>& a, bool invert) {
    int n = a.size();
    for (int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1) j ^= bit;
        j ^= bit;
        if (i < j) swap(a[i], a[j]);
    }
    for (int len = 2; len <= n; len <= 1) {
        double ang = 2 * acos(-1.0) / len * (invert ?
        -1 : 1);
        cd wlen(cos(ang), sin(ang));
        for (int i = 0; i < n; i += len) {
            cd w(1);
            for (int j = 0; j < len / 2; j++) {
                cd u = a[i + j], v = a[i + j + len / 2] *
                w;
                a[i + j] = u + v; a[i + j + len / 2] = u
                - v; w *= wlen;
            }
        }
        if (invert) for (cd& x : a) x /= n;
    }
}

// Multiplicacion de polinomios (coeficientes
// enteros)  $O(n \log n)$ 
vector<int> multiply(vector<int> const& a, vector
<int> const& b) {
    vector<cd> fa(all(a)), fb(all(b));
    int n = 1;
    while (n < a.size() + b.size()) n <= 1;
    fa.resize(n); fb.resize(n);
    fft(fa, false); fft(fb, false);
    for (int i = 0; i < n; i++) fa[i] *= fb[i];
    fft(fa, true);
    vector<int> res(n);
    for (int i = 0; i < n; i++) res[i] = round(fa[i]
    ].real());
    while (res.size() > 1 && !res.back()) res.
    pop_back();
    return res;
}

```

```

/**
 * Uso: ntt(vec, invert); auto c = multiply_mod(a
 * , b);
 * Transformada numerica rapida (NTT) y
 * multiplicacion de polinomios con coeficientes
 * modulares.
 * Complejidad:  $O(N \log N)$  con  $N$  potencia de dos.
 */
void ntt(vector<ll>& a, bool invert) {
    int n = a.size();
    for (int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1) j ^= bit;
        j ^= bit;
        if (i < j) swap(a[i], a[j]);
    }
    for (int len = 2; len <= n; len <= 1) {
        ll wlen = modpow(3, (MOD - 1) / len);
        if (invert) wlen = modinv(wlen);
        for (int i = 0; i < n; i += len) {
            ll w = 1;
            for (int j = 0; j < len / 2; j++) {
                ll u = a[i + j], v = a[i + j + len / 2] *
                w % MOD;
                a[i + j] = (u + v) % MOD;
                a[i + j + len / 2] = (u - v + MOD) % MOD;
                w = w * wlen % MOD;
            }
        }
        if (invert) {
            ll ninv = modinv(n);
            for (ll& x : a) x = x * ninv % MOD;
        }
    }
}

vector<ll> multiply_mod(vector<ll> const& a,
vector<ll> const& b) {
    vector<ll> fa(all(a)), fb(all(b));
    int n = 1;
    while (n < a.size() + b.size()) n <= 1;
    fa.resize(n); fb.resize(n);
    ntt(fa, false); ntt(fb, false);
    for (int i = 0; i < n; i++) fa[i] = fa[i] * fb[i]
    % MOD;
    ntt(fa, true);
    while (fa.size() > 1 && !fa.back()) fa.pop_back
    ();
    return fa;
}

/**
 * Uso: nim_game(pilas), sum_n(n), sum_squares(n)
 * , geom_sum(a, r, n);
 * Decide el ganador del juego de Nim y calcula
 * sumatorias cerradas (aritmética, cuadrados y
 * geométrica) modulo  $M$ .
 * Complejidad:  $O(N)$  para Nim,  $O(\log M)$  para las
 * sumatorias.
 */
bool nim_game(const vi& pilas) {
    int x = 0;
    for (int p : pilas) x ^= p;
    return x != 0;
}

ll sum_n(ll n, ll m = MOD) { n %= m; return n * (

```

```

    n + 1) % m * modinv(2, m) % m; }
ll sum_squares(ll n, ll m = MOD) {
    n %= m;
    return n * (n + 1) % m * (2 * n + 1) % m *
        modinv(6, m) % m;
}
ll geom_sum(ll a, ll r, ll n, ll m = MOD) {
    if (r == 1) return (n % m) * (a % m) % m;
    ll num = (modpow(r, n, m) - 1 + m) % m;
    return (a % m) * num % m * modinv((r - 1 + m) %
        m, m) % m;
}

```

```

/**
 * Uso: Matrix m(r, c); m.mat; m * o; m.power(p);
 * Matrix::identity(n);
 * Matriz con multiplicacion modular y
 * exponenciacion para recurrencias lineales y
 * conteo de caminos.
 * Complejidad: producto  $O(r \cdot c^2)$ , potencia
 *  $O(r^3 \log p)$ .
 */
struct Matrix {
    int r, c;
    vector<vector<ll>> mat;
    Matrix(int r, int c) : r(r), c(c), mat(r,
        vector<ll>(c, 0)) {}
    static Matrix identity(int n) {
        Matrix I(n, n);
        for (int i = 0; i < n; i++) I.mat[i][i] = 1;
        return I;
    }
    Matrix operator*(const Matrix& o) const {
        Matrix res(r, o.c);
        for (int i = 0; i < r; i++)
            for (int k = 0; k < c; k++) if (mat[i][k])
                for (int j = 0; j < o.c; j++)
                    res.mat[i][j] = (res.mat[i][j] + mat[i]
                        ][k] * o.mat[k][j]) % MOD;
        return res;
    }
    Matrix power(ll p) const {
        Matrix res = identity(r), base = *this;
        while (p) { if (p & 1) res = res * base; base =
            base * base; p >>= 1; }
        return res;
    }
};

```

```

/**
 * Uso: auto C = berlekamp_massey(sequencia);
 * Encuentra la recurrencia lineal minima de una
 * secuencia;  $C[0] \cdot S[n] + C[1] \cdot S[n-1] + \dots = 0$ .
 * Complejidad:  $O(N^2)$ .
 */
vector<ll> berlekamp_massey(const vector<ll>& s)
{
    vector<ll> C{1}, B{1};
    int L = 0; ll m = 1, b = 1;
    for (int i = 0; i < s.size(); i++) {
        ll d = 0;
        for (int j = 0; j <= L; j++) d = (d + C[j] *
            s[i - j]) % MOD;
        if (d) {
            vector<ll> T = C;
            ll coef = MOD - d * modpow(b, MOD - 2) %
                MOD;

```

```

            while (C.size() <= B.size() + m) C.pb(0);
            for (int j = 0; j < B.size(); j++) C[j + m]
                = (C[j + m] + coef * B[j]) % MOD;
            if (2 * L <= i) L = i + 1 - L, B = T, b = d
                , m = 1;
            else m++;
        } else m++;
    }
    C.resize(L + 1);
    return C;
}

```

```

/**
 * Uso: PollardRho::is_prime(n); PollardRho::
 * factorize(n);
 * Test de primalidad Miller-Rabin y
 * factorizacion de enteros de hasta  $2^{64}$  con
 * Pollard-Rho.
 * Complejidad:  $O(n^{1/4})$  esperado por factor
 * encontrado.
 */
struct PollardRho {
    using u64 = uint64_t; using u128 = __uint128_t;
    static u64 modpow(u64 b, u64 e, u64 m) {
        u64 r = 1; b %= m;
        while (e) { if (e & 1) r = (u128)r * b % m; b =
            (u128)b * b % m; e >>= 1; }
        return r;
    }
    static bool is_prime(u64 n) {
        if (n < 2) return false;
        for (u64 p : {2, 3, 5, 7, 11, 13, 17, 19, 23,
            29, 31, 37}) {
            if (n == p) return true;
            if (n % p == 0) return false;
        }
        u64 d = n - 1; int s = 0;
        while (!(d & 1)) d >>= 1, s++;
        for (u64 a : {2, 3, 5, 7, 11, 13, 17, 19, 23,
            29, 31, 37}) {
            if (a >= n) continue;
            u64 x = modpow(a, d, n);
            if (x == 1 || x == n - 1) continue;
            bool comp = true;
            for (int r = 1; r < s; r++) {
                x = (u128)x * x % n;
                if (x == n - 1) { comp = false; break; }
            }
            if (comp) return false;
        }
        return true;
    }
    static u64 get_factor(u64 n) {
        if (!(n & 1)) return 2;
        if (is_prime(n)) return n;
        u64 x = 2, y = 2, d = 1, c = 1;
        auto f = [&](u64 x) { return (u64)(((u128)x *
            x % n + c) % n); };
        while (d == 1) {
            x = f(x); y = f(f(y));
            d = gcd(x > y ? x - y : y - x, n);
            if (d == n) { x = rand() % (n - 2) + 2; y =
                x; c = rand() % (n - 1) + 1; d = 1; }
        }
        return d;
    }
}

```

```

static void factor(u64 n, map<u64, int>& f) {
    if (n == 1) return;
    if (is_prime(n)) { f[n]++; return; }
    u64 d = get_factor(n);
    factor(d, f); factor(n / d, f);
}

static map<u64, int> factorize(u64 n) { // {
    primo, exponente}
    map<u64, int> f;
    factor(n, f);
    return f;
};

/**
 * Uso: Lucas l(p); l.solve(n, r);
 * Combinaciones modulo un primo pequeno con n y
 * r potencialmente enormes (teorema de Lucas).
 * Complejidad: O(p) de preproceso, O(logp n) por
 * consulta.
 */
struct Lucas {
    Combinatorics comb;
    ll P;
    Lucas(ll p) : P(p), comb(p - 1, p) {}
    ll solve(ll n, ll r) {
        if (r < 0 || r > n) return 0;
        if (!r) return 1;
        if (n % P < r % P) return 0;
        return solve(n / P, r / P) * comb.nCr(n % P,
            r % P) % P;
    }
};

/**
 * Uso: fwht_xor(v, invert); auto c =
 * xor_convolution(a, b); // igual para or_ y and_
 * Transformada de Walsh-Hadamard y convoluciones
 * sobre operaciones de bits (xor, and, or)
 * modulo M.
 * Complejidad: O(N log N).
 */
void fwht_xor(vector<ll>& a, bool invert) {
    int n = a.size();
    for (int len = 2; len <= n; len <= 1) {
        for (int i = 0; i < n; i += len) {
            for (int j = 0; j < len / 2; j++) {
                ll u = a[i + j], v = a[i + j + len / 2];
                a[i + j] = (u + v) % MOD;
                a[i + j + len / 2] = (u - v + MOD) % MOD;
            }
        }
    }
    if (invert) {
        ll inv = modinv(n);
        for (ll& x : a) x = x * inv % MOD;
    }
}

void fwht_or(vector<ll>& a, bool invert) {
    int n = a.size();
    for (int len = 2; len <= n; len <= 1) {
        for (int i = 0; i < n; i += len) {
            for (int j = 0; j < len / 2; j++) {
                ll u = a[i + j], v = a[i + j + len / 2];
                if (invert) a[i + j + len / 2] = (v - u +
                    MOD) % MOD;
                else a[i + j + len / 2] = (u + v) % MOD;
            }
        }
    }
}

void fwht_and(vector<ll>& a, bool invert) {
    int n = a.size();
    for (int len = 2; len <= n; len <= 1) {
        for (int i = 0; i < n; i += len) {
            for (int j = 0; j < len / 2; j++) {
                ll u = a[i + j], v = a[i + j + len / 2];
                if (invert) a[i + j] = (u - v + MOD) %
                    MOD;
                else a[i + j] = (u + v) % MOD;
            }
        }
    }
}

vector<ll> xor_convolution(vector<ll> a, vector<
    ll> b) {
    int n = 1;
    while (n < max(a.size(), b.size())) n <= 1;
    a.resize(n); b.resize(n);
    fwht_xor(a, false); fwht_xor(b, false);
    for (int i = 0; i < n; i++) a[i] = a[i] * b[i]
        % MOD;
    fwht_xor(a, true);
    return a;
}

vector<ll> or_convolution(vector<ll> a, vector<ll>
    > b) {
    int n = 1;
    while (n < max(a.size(), b.size())) n <= 1;
    a.resize(n); b.resize(n);
    fwht_or(a, false); fwht_or(b, false);
    for (int i = 0; i < n; i++) a[i] = a[i] * b[i]
        % MOD;
    fwht_or(a, true);
    return a;
}

vector<ll> and_convolution(vector<ll> a, vector<
    ll> b) {
    int n = 1;
    while (n < max(a.size(), b.size())) n <= 1;
    a.resize(n); b.resize(n);
    fwht_and(a, false); fwht_and(b, false);
    for (int i = 0; i < n; i++) a[i] = a[i] * b[i]
        % MOD;
    fwht_and(a, true);
    return a;
}

/**
 * Uso: floor_sum(n, m, a, b);
 * Calcula  $\sum_{i=0}^{n-1} \lfloor (a \cdot i + b) / m \rfloor$ .
 * Complejidad: O(log m).
 */
ll floor_sum(ll n, ll m, ll a, ll b) {
    ll ans = 0;
    while (true) {
        if (a >= m) { ans += (n - 1) * n * (a / m) /
            2; a %= m; }
        if (b >= m) { ans += n * (b / m); b %= m; }
        ll y_max = a * n + b;
        if (y_max < m) break;
        n = y_max / m;
    }
}

```

```

    b = y_max % m;
    swap(m, a);
}
return ans;
}
}

```

Geometry

```

/**
 * Uso: cmp(a, b);
 * Comparacion de numeros de punto flotante con
 * tolerancia EPS; devuelve -1, 0 o 1.
 * Complejidad: O(1).
 */
int cmp(ld x, ld y = 0) { return x <= y + EPS ? (x
+ EPS < y ? -1 : 0) : 1; }

/**
 * Uso: Point<ll> p(x, y); p+q, p-q, p*c, p/c, p==q
 * ; orden lexicografico con <
 * Punto 2D generico con operadores aritmeticos,
 * igualdad y orden lexicografico.
 * Complejidad: O(1) por operacion.
 */
template<typename T> struct Point {
    T x, y;
    Point(T x = 0, T y = 0) : x(x), y(y) {}
    Point operator+(const Point& p) const { return {x
+ p.x, y + p.y}; }
    Point operator-(const Point& p) const { return {x
- p.x, y - p.y}; }
    Point operator*(T c) const { return {x * c, y * c
}; }
    Point operator/(T c) const { return {x / c, y / c
}; }
    bool operator==(const Point& p) const { return x
== p.x && y == p.y; }
    bool operator<(const Point& p) const { return x <
p.x || (x == p.x && y < p.y); } //
    lexicografico
};

/**
 * Uso: dot(a, b), cross(a, b), cross(p, a, b),
 * norm2(p), norm(p), sgn(v);
 * Operaciones vectoriales basicas: producto punto,
 * producto cruz, norma al cuadrado, norma y
 * signo.
 * Complejidad: O(1) por operacion.
 */
template<typename T> T dot(Point<T> a, Point<T> b)
{ return a.x * b.x + a.y * b.y; }
template<typename T> T cross(Point<T> a, Point<T> b
) { return a.x * b.y - a.y * b.x; }
template<typename T> T cross(Point<T> p, Point<T> a
, Point<T> b) { return cross(a - p, b - p); }
template<typename T> T norm2(Point<T> p) { return
dot(p, p); }
template<typename T> ld norm(Point<T> p) { return
sqrt1(norm2(p)); }
template<typename T> int sgn(T v) { return (T(0) <
v) - (v < T(0)); }

/**
 * Uso: orient(a, b, c), on_segment(p, a, b),
 * segment_intersect(a, b, c, d);
 * Orientacion de tres puntos (horaria, colineal o
 * antihoraria), pertenencia de un punto a un
 * segmento e interseccion de segmentos.

```

```

 * Complejidad: O(1) por operacion.
 */
template<typename T> int orient(Point<T> a, Point<T>
> b, Point<T> c) { return sgn(cross(a, b, c)); }

template<typename T> bool on_segment(Point<T> p,
Point<T> a, Point<T> b) {
    return orient(a, b, p) == 0 && min(a.x, b.x) <= p
.x && p.x <= max(a.x, b.x) &&
        min(a.y, b.y) <= p.y && p.y <= max(a.y, b
.y);
}

template<typename T> bool segment_intersect(Point<T>
> a, Point<T> b, Point<T> c, Point<T> d) {
    int o1 = orient(a, b, c), o2 = orient(a, b, d),
    o3 = orient(c, d, a), o4 = orient(c, d, b);
    if (o1 != o2 && o3 != o4) return true;
    if (o1 == 0 && on_segment(c, a, b)) return true;
    if (o2 == 0 && on_segment(d, a, b)) return true;
    if (o3 == 0 && on_segment(a, c, d)) return true;
    if (o4 == 0 && on_segment(b, c, d)) return true;
    return false;
}

/**
 * Uso: polygon_area_2(poly) / 2 = area,
 * boundary_points(poly), point_in_polygon(poly, p
);
 * Area doble de un poligono (para evitar floats),
 * puntos del reticulo en el borde (teorema de
 * Pick) y clasificacion de un punto (0 fuera, 1
 * borde, 2 dentro).
 * Complejidad: O(N) por operacion con N vertices.
 */
template<typename T> T polygon_area_2(const vector<
Point<T>&& p) {
    T area = 0;
    for (int i = 0; i < p.size(); i++) area += cross(
p[i], p[(i + 1) % p.size()]);
    return abs(area);
}

// Puntos del reticulo en el borde (Pick: 2*A = 2*I
+ B - 2)
long long boundary_points(const vector<Point<ll>&&
poly) {
    ll b = 0;
    for (int i = 0; i < poly.size(); i++) {
        auto p1 = poly[i], p2 = poly[(i + 1) % poly
.size()];
        b += gcd(abs(p1.x - p2.x), abs(p1.y - p2.y));
    }
    return b;
}

// Retorna 0 afuera, 1 en el borde, 2 adentro
template<typename T> int point_in_polygon(const
vector<Point<T>&& poly, Point<T> p) {
    int hits = 0;
    for (int i = 0; i < poly.size(); i++) {
        auto a = poly[i], b = poly[(i + 1) % poly.size
()];
        if (on_segment(p, a, b)) return 1;
        if (a.y > b.y) swap(a, b);
        if (p.y >= a.y && p.y < b.y && cross(p, a, b) >
0) hits++;
    }
}

```

```

    }
    return hits & 1 ? 2 : 0;
}

/**
 * Uso: auto hull = convex_hull(puntos);
 * Cascara convexa (monotonic chain) en sentido
 * antihorario, preservando puntos colineales en
 * el borde.
 * Complejidad:  $O(N \log N)$ .
 */
template<typename T> vector<Point<T>> convex_hull(
    vector<Point<T>> pts) {
    int n = pts.size(), k = 0;
    if (n <= 2) return pts;
    vector<Point<T>> h(2 * n);
    sort(all(pts));
    for (int i = 0; i < n; i++) {
        while (k >= 2 && cross(h[k - 2], h[k - 1], pts[i]) < 0) k--;
        h[k++] = pts[i];
    }
    for (int i = n - 2, t = k + 1; i >= 0; i--) {
        while (k >= t && cross(h[k - 2], h[k - 1], pts[i]) < 0) k--;
        h[k++] = pts[i];
    }
    h.resize(k - 1);
    return h;
}

/**
 * Uso: closest_pair(puntos);
 * Distancia al cuadrado entre el par de puntos mas
 * cercanos.
 * Complejidad:  $O(N \log N)$ .
 */
long long closest_pair(vector<Point<ll>> pts) {
    int n = pts.size();
    if (n < 2) return 8e18;
    sort(all(pts));
    set<pair<ll, ll>> active; // {y, x}
    ll best = 8e18;
    int left = 0;
    for (int i = 0; i < n; i++) {
        ll d = ceil(sqrt(1ll * best));
        while (left < i && pts[i].x - pts[left].x >= d)
            active.erase({pts[left].y, pts[left].x}), left++;
        auto lo = active.lower_bound({pts[i].y - d, -8e18});
        auto hi = active.upper_bound({pts[i].y + d, 8e18});
        for (auto it = lo; it != hi; ++it) {
            ll dy = pts[i].y - it->fi, dx = pts[i].x - it->se;
            best = min(best, dx * dx + dy * dy);
        }
        active.insert({pts[i].y, pts[i].x});
    }
    return best;
}

/**
 * Uso: Event1D / Event2D para ordenar eventos de
 * sweep line por coordenada x.
 * Eventos para sweep line en 1D y 2D; type +1

```

```

    entrada/inicio, -1 salida/fin.
 * Complejidad:  $O(1)$  por evento,  $O(K \log K)$  al
    ordenarlos con  $K$  eventos.
 */
struct Event1D {
    ll x;
    int type, id;
    bool operator<(const Event1D& o) const { return x
        != o.x ? x < o.x : type > o.type; }
};
struct Event2D {
    ll x, y1, y2;
    int type; // +1 borde izq, -1 borde der
    bool operator<(const Event2D& o) const { return x
        != o.x ? x < o.x : type > o.type; }
};

/**
 * Uso: polar_sort(puntos, centro);
 * Ordena puntos por angulo polar alrededor de un
 * centro, en sentido antihorario desde el eje +x.
 * Complejidad:  $O(N \log N)$ .
 */
template<typename T> void polar_sort(vector<Point<T>
    >& pts, Point<T> center = {}) {
    auto half = [](Point<T> p) { return p.y > 0 || (p
        .y == 0 && p.x > 0) ? 1 : -1; };
    sort(all(pts), [&](Point<T> a, Point<T> b) {
        auto va = a - center, vb = b - center;
        int ha = half(va), hb = half(vb);
        if (ha != hb) return ha > hb;
        T cr = cross(va, vb);
        if (cr != 0) return cr > 0;
        return norm2(va) < norm2(vb);
    });
}

/**
 * Uso: dist_to_line(p, a, b), dist_to_segment(p, a
    , b), line_intersect(a, b, c, d);
 * Distancias de un punto a una linea o segmento e
    interseccion exacta de dos lineas infinitas no
    paralelas.
 * Complejidad:  $O(1)$  por operacion.
 */
template<typename T> ld dist_to_line(Point<T> p,
    Point<T> a, Point<T> b) {
    return abs((ld)cross(p, a, b)) / norm(a - b);
}
template<typename T> ld dist_to_segment(Point<T> p,
    Point<T> a, Point<T> b) {
    if (a == b) return norm(p - a);
    if (dot(p - a, b - a) <= 0) return norm(p - a);
    if (dot(p - b, a - b) <= 0) return norm(p - b);
    return dist_to_line(p, a, b);
}

// Interseccion exacta de lineas infinitas AB y CD (
    no paralelas)
Point<ld> line_intersect(Point<ld> a, Point<ld> b,
    Point<ld> c, Point<ld> d) {
    ld cr1 = cross(b - a, c - a), cr2 = cross(b - a,
        d - a);
    return c + (d - c) * (cr1 / (cr1 - cr2));
}

/**

```

```

* Uso: rotating_calipers(hull);
* Diametro al cuadrado de un poligono convexo
* mediante calipers rotativos.
* Complejidad:  $O(N)$ .
*/
template<typename T> T rotating_calipers(const
    vector<Point<T>>& poly) {
    int n = poly.size();
    if (n < 2) return 0;
    if (n == 2) return norm2(poly[0] - poly[1]);
    T best = 0;
    int j = 1;
    for (int i = 0; i < n; i++) {
        int ni = (i + 1) % n;
        while (abs(cross(poly[j], poly[i], poly[ni])) <
            abs(cross(poly[(j + 1) % n], poly[i], poly[ni]
            )))) j = (j + 1) % n;
        best = max({best, norm2(poly[i] - poly[j]),
            norm2(poly[ni] - poly[j])});
    }
    return best;
}

/**
* Uso: Circle c(centro, radio);
* circle_line_intersect(a, b, c);
* circle_circle_intersect(c1, c2);
* Circulo e intersecciones con lineas y otros
* circulos; devuelven 0, 1 o 2 puntos.
* Complejidad:  $O(1)$  por operacion.
*/
struct Circle {
    Point<ld> c;
    ld r;
    Circle(Point<ld> c = {}, ld r = 0) : c(c), r(r)
    {}
};

vector<Point<ld>> circle_line_intersect(Point<ld> a
    , Point<ld> b, Circle circ) {
    auto dir = b - a;
    ld len2 = norm2(dir);
    if (cmp(len2) == 0) return {};
    ld t = dot(circ.c - a, dir) / len2;
    auto proj = a + dir * t;
    ld d2 = norm2(circ.c - proj);
    if (cmp(d2, circ.r * circ.r) > 0) return {};
    if (cmp(d2, circ.r * circ.r) == 0) return {proj};
    ld off = sqrt1(max((ld)0, circ.r * circ.r - d2) /
        len2);
    return {proj - dir * off, proj + dir * off};
}

vector<Point<ld>> circle_circle_intersect(Circle c1
    , Circle c2) {
    auto dir = c2.c - c1.c;
    ld d = norm(dir);
    if (cmp(d, c1.r + c2.r) > 0 || cmp(d, abs(c1.r -
        c2.r)) < 0 || cmp(d) == 0) return {};
    ld a = (c1.r * c1.r - c2.r * c2.r + d * d) / (2 *
        d);
    ld h = sqrt1(max((ld)0, c1.r * c1.r - a * a));
    auto p2 = c1.c + dir * (a / d);
    auto perp = Point<ld>(-dir.y, dir.x) * (h / d);
    if (cmp(h) == 0) return {p2};
    return {p2 + perp, p2 - perp};
}

```

```

/**
* Uso: vector<Halfplane> H; H.pb(Halfplane(a, b));
* halfplane_intersection(H);
* Interseccion de semiplanos (valida a la
* izquierda de cada recta  $a \rightarrow b$ ); devuelve el
* poligono convexo resultante.
* Complejidad:  $O(N \log N)$ .
*/
struct Halfplane {
    Point<ld> p, pq; // p = punto, pq = vector
    // direccional
    ld angle;
    Halfplane() {}
    Halfplane(Point<ld> a, Point<ld> b) : p(a), pq(b
        - a), angle(atan2(pq.y, pq.x)) {}
    bool out(Point<ld> r) const { return cross(pq, r
        - p) < -EPS; } // r afuera (lado derecho
    // estricto)
    bool operator<(const Halfplane& e) const {
        if (abs(angle - e.angle) > EPS) return angle <
            e.angle;
        return cross(pq, e.p - p) > EPS;
    }
};

Point<ld> intersect(Halfplane s, Halfplane t) {
    ld alpha = cross(t.p - s.p, t.pq) / cross(s.pq, t
        .pq);
    return s.p + s.pq * alpha;
}

vector<Point<ld>> halfplane_intersection(vector<
    Halfplane>& H) {
    sort(all(H));
    int n = H.size(), k = 0;
    vector<Halfplane> U(n);
    for (int i = 0; i < n; i++) {
        if (i > 0 && abs(H[i].angle - H[i - 1].angle) <
            EPS) continue;
        U[k++] = H[i];
    }
    H = U; H.resize(k); n = k;
    deque<Halfplane> dq;
    dq.push_back(H[0]);
    if (n > 1) dq.push_back(H[1]);
    for (int i = 2; i < n; i++) {
        while (dq.size() >= 2 && H[i].out(intersect(dq.
            back(), dq[dq.size() - 2]))) dq.pop_back();
        while (dq.size() >= 2 && H[i].out(intersect(dq.
            front(), dq[1]))) dq.pop_front();
        dq.push_back(H[i]);
    }
    while (dq.size() >= 3 && dq.front().out(intersect
        (dq.back(), dq[dq.size() - 2]))) dq.pop_back();
    while (dq.size() >= 3 && dq.back().out(intersect(
        dq.front(), dq[1]))) dq.pop_front();
    vector<Point<ld>> poly;
    if (dq.size() < 3) return poly;
    for (int i = 0; i < dq.size(); i++) poly.pb(
        intersect(dq[i], dq[(i + 1) % dq.size()]));
    vector<Point<ld>> clean; // quita vertices
    // duplicados y colineales
    for (int i = 0; i < poly.size(); i++) {
        auto cur = poly[i], nxt = poly[(i + 1) % poly.
            size()];
        if (hypot(cur.x - nxt.x, cur.y - nxt.y) < EPS)
            continue;
    }
}

```

```

    if (!clean.empty()) {
        auto prv = clean.back();
        if (abs(cross(cur - prv, nxt - cur)) < EPS)
            continue;
    }
    clean.pb(cur);
}
return clean;
}

/**
 * Uso: Point3D p(x, y, z); dist_point_plane(p, a,
 * n); dist_point_line(p, a, b);
 * line_plane_intersect(L1, L2, P, N);
 * Punto 3D con producto punto y cruz, distancias
 * punto-plano y punto-linea e interseccion linea-
 * plano.
 * Complejidad: O(1) por operacion.
 */
struct Point3D {
    double x, y, z;
    Point3D(double x = 0, double y = 0, double z = 0)
        : x(x), y(y), z(z) {}
    Point3D operator+(const Point3D& p) const {
        return {x + p.x, y + p.y, z + p.z};
    }
    Point3D operator-(const Point3D& p) const {
        return {x - p.x, y - p.y, z - p.z};
    }
    Point3D operator*(double c) const { return {x * c,
        y * c, z * c}; }
    double dot(const Point3D& p) const { return x * p
        .x + y * p.y + z * p.z; }
    Point3D cross(const Point3D& p) const { return {y
        * p.z - z * p.y, z * p.x - x * p.z, x * p.y -
        y * p.x}; }
    double length() const { return sqrt(dot(*this));
    }
};

double dist_point_plane(Point3D p, Point3D a,
    Point3D n) { return abs((p - a).dot(n)) / n.
    length(); }

double dist_point_line(Point3D p, Point3D a,
    Point3D b) { return (p - a).cross(b - a).length
    () / (b - a).length(); }

Point3D line_plane_intersect(Point3D L1, Point3D L2
    , Point3D P, Point3D N) {
    auto u = L2 - L1;
    double d = N.dot(u);
    if (abs(d) > EPS) return L1 + u * ((P - L1).dot(N)
        ) / d;
    return {1e9, 1e9, 1e9}; // paralelos
}

```

Bit Operations

```

/**
 * Uso: check_bit(m, i), set_bit(m, i), clear_bit(m
 * , i), toggle_bit(m, i), lsb(m). i 0-based.
 * Manipulacion basica de bits de una mascara:
 * consultar, activar, limpiar, alternar y aislar
 * el bit menos significativo.
 * Complejidad: O(1) por operacion.
 */
inline bool check_bit(ll mask, int i) { return (
    mask >> i) & 1; }
inline ll set_bit(ll mask, int i) { return mask |
    (1LL << i); }

```

```

inline ll clear_bit(ll mask, int i) { return mask &
    ~(1LL << i); }
inline ll toggle_bit(ll mask, int i) { return mask
    ^ (1LL << i); }
inline ll lsb(ll mask) { return mask & -mask; }
/**
 * Uso: count_bits(x), count_bits_ll(x),
 * highest_bit(x) = bit mas alto (-1 si 0), ctz(x)
 * = ceros a la derecha.
 * Conteo de bits activos, posicion del bit mas
 * alto y numero de ceros finales con instrucciones
 * nativas.
 * Complejidad: O(1) por operacion.
 */
inline int count_bits(int mask) { return
    __builtin_popcount(mask); }
inline int count_bits_ll(ll mask) { return
    __builtin_popcountll(mask); }
inline int highest_bit(ll mask) { return mask == 0
    ? -1 : 63 - __builtin_clzll(mask); }
inline int ctz(ll mask) { return mask == 0 ? 64 :
    __builtin_ctzll(mask); }

/**
 * Uso: iterate_submasks(m);
 * Itera todas las sub-mascaras de una mascara dada
 * en orden decreciente (sin incluir la vacia).
 * Complejidad: O(3N) en total para una mascara de
 * N bits.
 */
inline void iterate_submasks(ll mask) {
    for (ll sub = mask; sub; sub = (sub - 1) & mask)
        { /* procesar sub */ }
}

/**
 * Uso: auto masks = gossers_hack(k, n);
 * Genera todas las mascarar de tamano n con
 * exactamente k bits activos (gossers hack).
 * Complejidad: O( $\binom{n}{k}$ ) mascarar generadas, O(1)
 * amortizado por mascara.
 */
inline vector<ll> gossers_hack(int k, int n) {
    vector<ll> res;
    if (!k) return {0};
    ll mask = (1LL << k) - 1, limit = 1LL << n;
    while (mask < limit) {
        res.pb(mask);
        ll c = mask & -mask, r = mask + c;
        mask = (((r ^ mask) >> 2) / c) | r;
    }
    return res;
}

/**
 * Uso: XorBasis b; b.insert(x); b.can_form(k); b.
 * get_max(init); b.get_min();
 * Base lineal de xor sobre enteros de 64 bits:
 * maximo xor de un subconjunto, minimo y test de
 * representabilidad.
 * Complejidad: insercion y consultas O(60).
 */
struct XorBasis {
    static const int BITS = 60;
    ll basis[BITS] = {};
    int sz = 0;
    // Inserta; true si fue linealmente independiente

```

```

bool insert(ll x) {
    for (int i = BITS - 1; i >= 0; i--) if (
        check_bit(x, i)) {
        if (!basis[i]) return basis[i] = x, sz++,
        true;
        x ^= basis[i];
    }
    return false;
}

bool can_form(ll k) {
    for (int i = BITS - 1; i >= 0; i--) if (
        check_bit(k, i)) {
        if (!basis[i]) return false;
        k ^= basis[i];
    }
    return true;
}

// Maximo de (initial ^ subset)
ll get_max(ll initial = 0) {
    for (int i = BITS - 1; i >= 0; i--) if ((
        initial ^ basis[i]) > initial) initial ^= basis
        [i];
    return initial;
}

ll get_min() {
    for (int i = 0; i < BITS; i++) if (basis[i])
        return basis[i];
    return 0;
}
};

/**
 * Uso: BitTrie t; t.insert(x); t.max_xor(x); t.
 *      insert(x, -1) borra.
 * Trie binario para consultar el maximo xor de un
 * valor contra el conjunto insertado, con soporte
 * de borrado.
 * Complejidad: insercion y consulta  $O(B)$  con  $B = 30$ 
 * bits (60 si los valores llegan a  $10^{18}$ ).
 */
struct BitTrie {
    static const int BITS = 30; // 60 si los numeros
    llegan a  $1e18$ 
    struct Node { int next[2] = {-1, -1}; int cnt =
    0; };
    vector<Node> t;
    BitTrie() { t.emplace_back(); }
    void insert(ll x, int val = 1) { // val=-1 para
    borrar
        int u = 0;
        for (int i = BITS - 1; i >= 0; i--) {
            int b = check_bit(x, i);
            if (t[u].next[b] == -1) t[u].next[b] = t.size
            (), t.emplace_back();
            u = t[u].next[b];
            t[u].cnt += val;
        }
    }
    ll max_xor(ll x) {
        int u = 0; ll ans = 0;
        for (int i = BITS - 1; i >= 0; i--) {
            int b = check_bit(x, i), opp = b ^ 1;
            int v = t[u].next[opp];
            if (v != -1 && t[v].cnt > 0) ans |= 1LL << i,
            u = v;
            else u = t[u].next[b];
        }
    }
};

```

```

    }
    return ans;
}
};

```

Extra

```

#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>

/**
 * Uso[0]: ordered_set<T> s; s.insert(x); s.
 *      order_of_key(x); *s.find_by_order(k);
 * Conjunto ordenado de pbds con consultas por
 * rango (k-esimo menor y posicion).
 * Complejidad:  $O(\log N)$  por operacion.
 */
template <typename T>
using ordered_set = __gnu_pbds::tree<T, __gnu_pbds
::null_type, less<T>, __gnu_pbds::rb_tree_tag,
__gnu_pbds::tree_order_statistics_node_update>;

/**
 * Uso[0]: ordered_multiset<T> s; s.insert({x, id})
 * ;
 * Conjunto ordenado de pbds que admite duplicados
 * emparejando cada valor con un id unico.
 * Complejidad:  $O(\log N)$  por operacion.
 */
template <typename T>
using ordered_multiset = __gnu_pbds::tree<pair<T,
int>, __gnu_pbds::null_type, less<pair<T, int
>>, __gnu_pbds::rb_tree_tag, __gnu_pbds::
tree_order_statistics_node_update>;

/**
 * Uso: unordered_map<ll, int, custom_hash> m;
 * Functor de hash anti-colision (splitmix64 con
 * semilla aleatoria) para tablas hash de claves
 * numericas.
 * Complejidad:  $O(1)$  por operacion de hash.
 */
struct custom_hash {
    static uint64_t splitmix64(uint64_t x) {
        x += 0x9e3779b97f4a7c15;
        x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
        x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
        return x ^ (x >> 31);
    }
    size_t operator()(uint64_t x) const {
        static const uint64_t FIXED_RANDOM = chrono::
        steady_clock::now().time_since_epoch().count();
        return splitmix64(x + FIXED_RANDOM);
    }
};

/**
 * Uso: MeetInTheMiddle::subset_sum(a, target);
 * Resuelve subset sum exacto con la tecnica meet
 * in the middle para arreglos de hasta 40
 * elementos.
 * Complejidad:  $O(N \cdot 2^{N/2})$  tiempo,  $O(2^{N/2})$  espacio.
 */
struct MeetInTheMiddle {
    static bool subset_sum(const vll& a, ll target) {
        int n = a.size(), mid = n / 2;
        vll left, right;
        for (int i = 0; i < (1 << mid); i++) {
            ll s = 0;
            for (int j = 0; j < mid; j++) if (i >> j & 1)
                s += a[j];
        }
    }
};

```

```

    left.pb(s);
}
int n2 = n - mid;
for (int i = 0; i < (1 << n2); i++) {
    ll s = 0;
    for (int j = 0; j < n2; j++) if (i >> j & 1)
        s += a[mid + j];
    right.pb(s);
}
sort(all(right));
for (ll s : left) if (binary_search(all(right),
    target - s)) return true;
return false;
}
};

/**
 * Uso[0]: ZobristHash zh; auto pref = zh.
 *           build_pref(a); zh.query_range(pref, l, r);
 * Hash de prefijos xor para comparar multiconjuntos
 * de rangos en  $O(1)$ .
 * Complejidad: construccion  $O(N)$ , consulta  $O(1)$ .
 */
struct ZobristHash {
    using u64 = uint64_t;
    mt19937_64 rng;
    unordered_map<int, u64> val_hash;
    ZobristHash(u64 seed = 1337) : rng(seed) {
        val_hash[0] = 0;
    }
    u64 get_hash(int x) {
        if (!val_hash.count(x)) val_hash[x] = rng();
        return val_hash[x];
    }
    vector<u64> build_pref(const vi& a) { // prefijos
        XOR
        vector<u64> pref(a.size() + 1, 0);
        for (int i = 0; i < a.size(); i++) pref[i + 1]
            = pref[i] ^ get_hash(a[i]);
        return pref;
    }
    u64 query_range(const vector<u64>& pref, int l,
        int r) { return pref[r + 1] ^ pref[l]; }
};

/**

```

```

 * Uso[1]: BitsetReachability<MAXN> br; br.build(n,
 *           aristas); br.can_reach(u, v);
 * Cierre transitivo (alcanzabilidad todos-pares)
 * de un grafo dirigido usando bitsets.
 * Complejidad: construccion  $O(N^3/64)$ , consulta  $O(1)$ 
 */
template<size_t MAXN>
struct BitsetReachability {
    bitset<MAXN> reachable[MAXN];
    void build(int n, const vector<pii>& edges) {
        for (int i = 1; i <= n; i++) reachable[i].reset();
        for (auto [u, v] : edges) reachable[u][v] = 1;
        for (int k = 1; k <= n; k++)
            for (int i = 1; i <= n; i++)
                if (reachable[i][k]) reachable[i] |=
                    reachable[k];
    }
    bool can_reach(int u, int v) { return reachable[u][v]; }
};

/**
 * Uso[0]: auto [vals, ids] = compress_coords(a);
 * Compresion de coordenadas: ordena y elimina
 * duplicados (sort + unique) y mapea cada valor a
 * su rango consecutivo  $0..K-1$ .
 * Indexación: ids 0-indexados en orden creciente
 * de valor.
 * Complejidad:  $O(N \log N)$  tiempo,  $O(N)$  espacio.
 */
template<typename T>
pair<vector<T>, vector<int>> compress_coords(const
    vector<T>& a) {
    vector<T> vals = a;
    sort(all(vals));
    vals.erase(unique(all(vals)), vals.end());
    vector<int> ids(a.size());
    for (int i = 0; i < (int)a.size(); i++)
        ids[i] = lower_bound(all(vals), a[i]) - vals.
            begin();
    return {vals, ids};
}

```